

中国科学院大学

University of Chinese Academy of Sciences

Generative Models & AI4S Applications

Lecture 11 Diffusion Models

Fall 2026

Liu Yunfan (刘云帆)

liuyunfan@ucas.ac.cn

Announcements

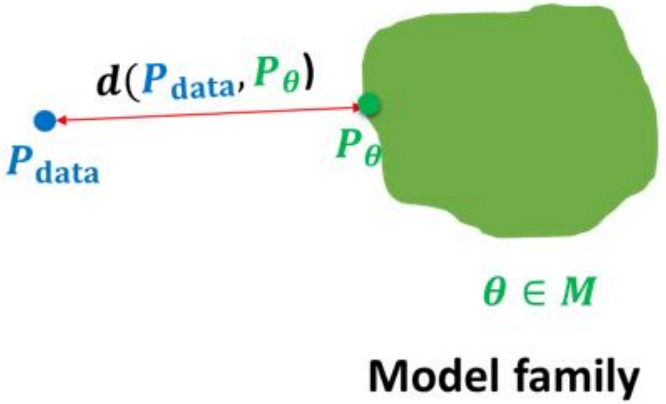
- Assignment 1 due on **Oct. 25th**
- Assignment 2 is released, and due on **Nov. 22th**
- This slides subject to future updates

Recap

Images of dogs



$x_i \sim P_{\text{data}}$
$i = 1, 2, \ldots, n$



Dog Image Generator

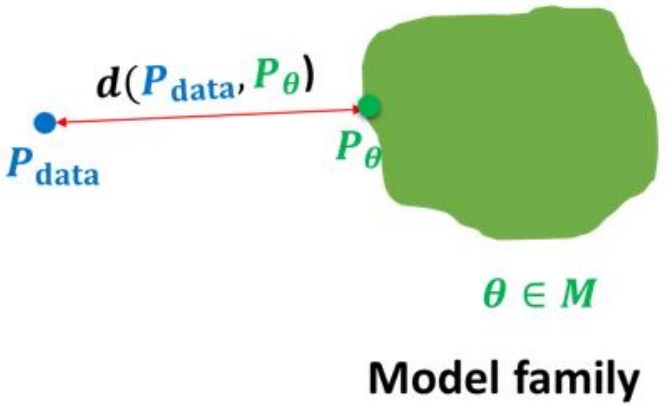
- Generative models learned via Maximum Likelihood:
  - **Autoregressive Models:** $p_\theta(x) = \prod_{i=1}^{n} p_\theta(x_i \mid x_{<i})$
  - **Variational Autoencoders:** $p_\theta(x) = \int p_\theta(x, z)\, dz$
  - **Normalizing Flow Models:** $p_X(x; \theta) = p_Z\left(f_\theta^{-1}(x)\right)\left|\det\left(\frac{\partial f_\theta^{-1}(x)}{\partial x}\right)\right|$
- **Explicitly** compute / approximate the density $p_\theta(x)$

Recap

Images of dogs



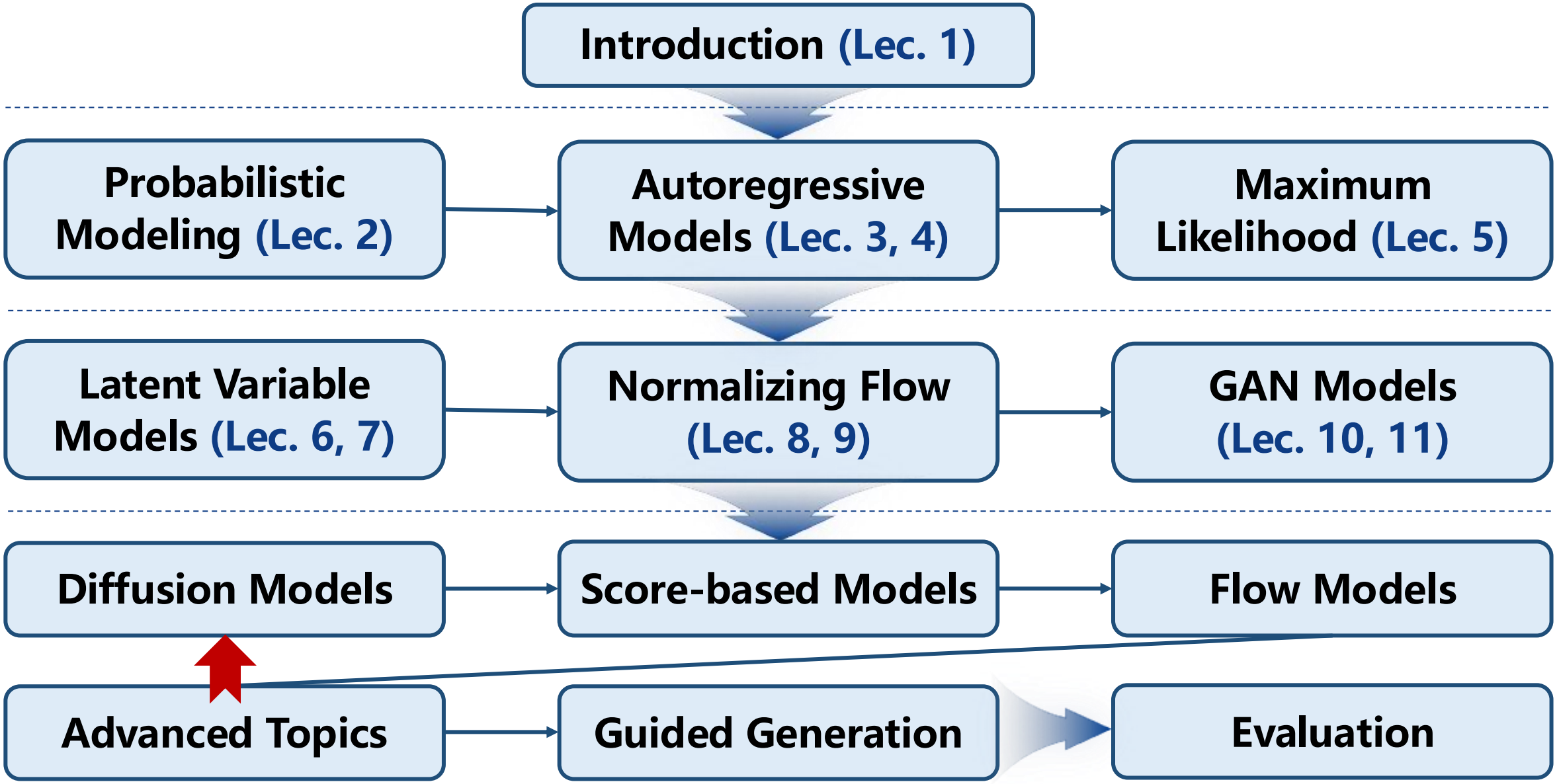
$x_i \sim P_{\text{data}}$
$i = 1, 2, \ldots, n$



Dog Image Generator

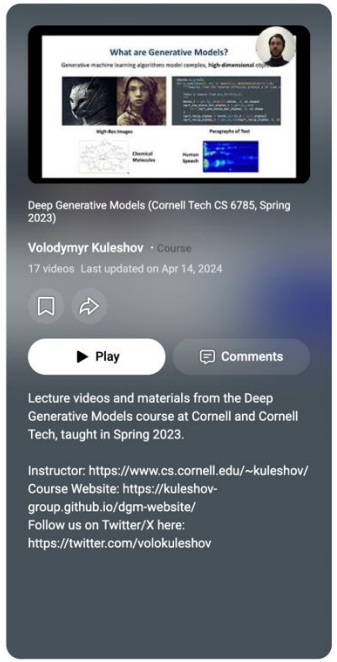
- Likelihood-free Learning: **GAN**
  - **Two-player minimax:** $\min_G \max_D V(G, D) = E_{\mathbf{x} \sim p_{\text{data}}}[\log D(\mathbf{x})] + E_{\mathbf{x} \sim p_G}[\log(1 - D(\mathbf{x}))]$
  - **Pros:** High-quality generation, flexible objective and architecture
  - **Cons:** Very unstable to train, suffer from mode collapse
- **Implicitly** induce the density $p(x)$ by the generator G_θ

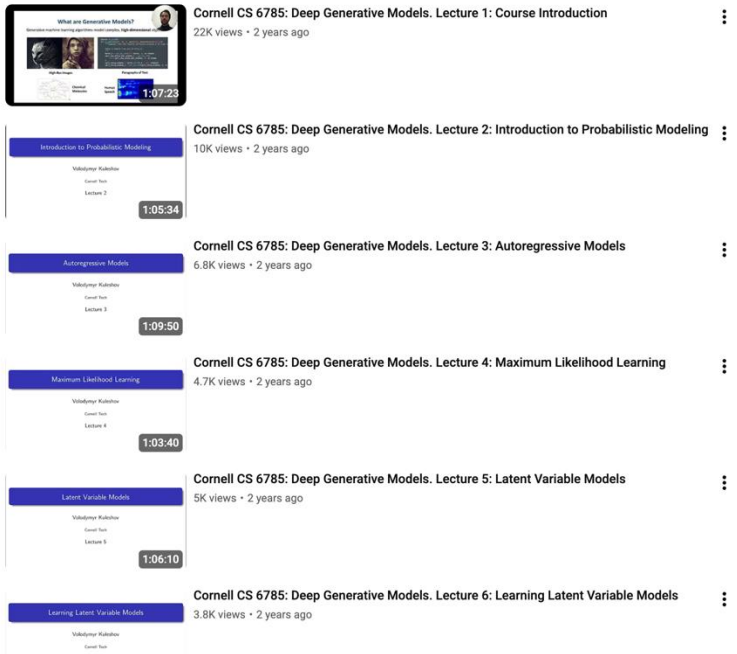
Roadmap

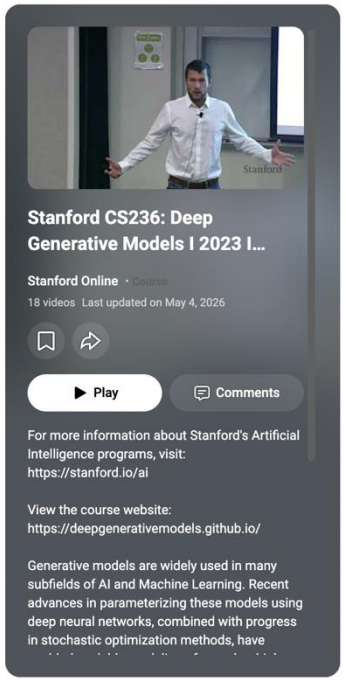

Introduction (Lec. 1)
Probabilistic Modeling (Lec. 2)
Autoregressive Models (Lec. 3, 4)
Maximum Likelihood (Lec. 5)
Latent Variable Models (Lec. 6, 7)
Normalizing Flow (Lec. 8, 9)
GAN Models (Lec. 10, 11)
Diffusion Models
Score-based Models
Flow Models
Advanced Topics
Guided Generation
Evaluation

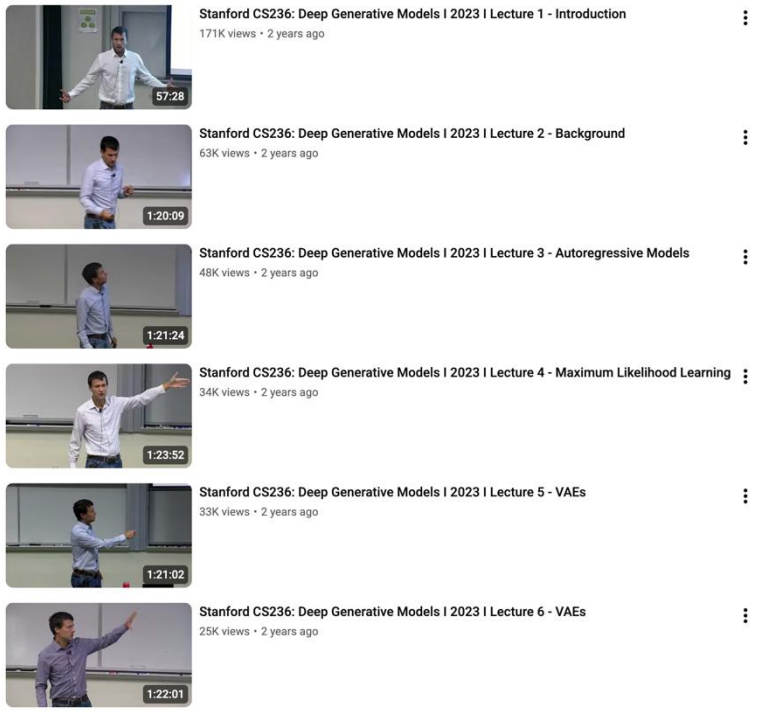
Plan for the Following Lectures

- **Diffusion and Flow-based Models**
  - ➢ Many fresh-new materials, featuring different introducing logistics







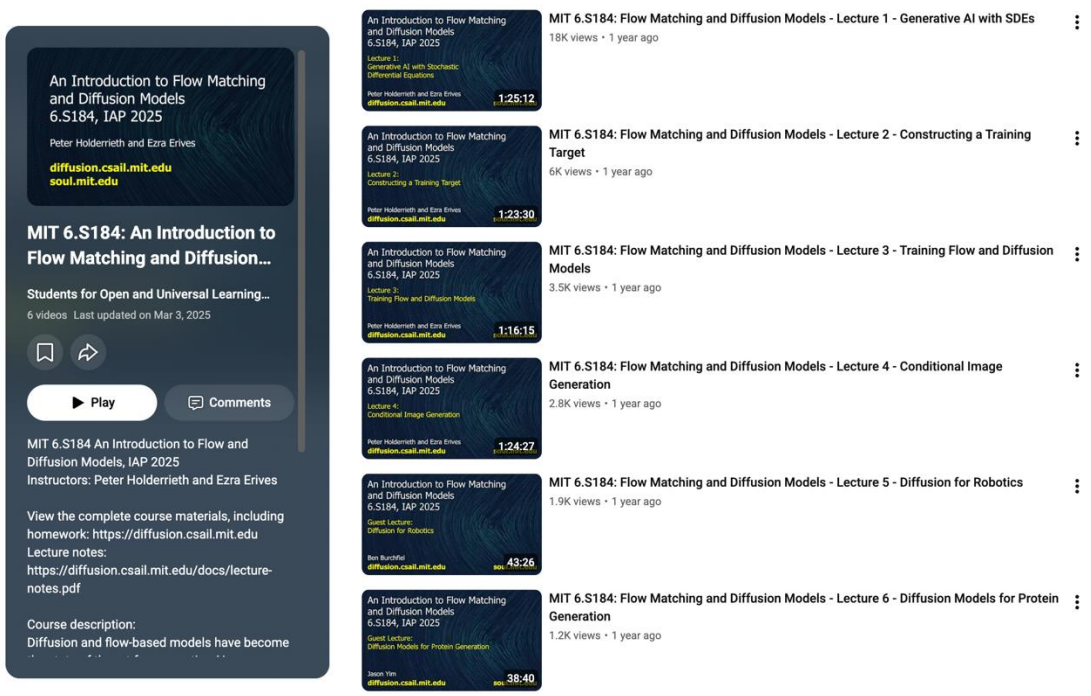


Stanford CS 236 / Cornell CS 6785

Energy-based Model ➔ Score-based Model ➔ Diffusion Model

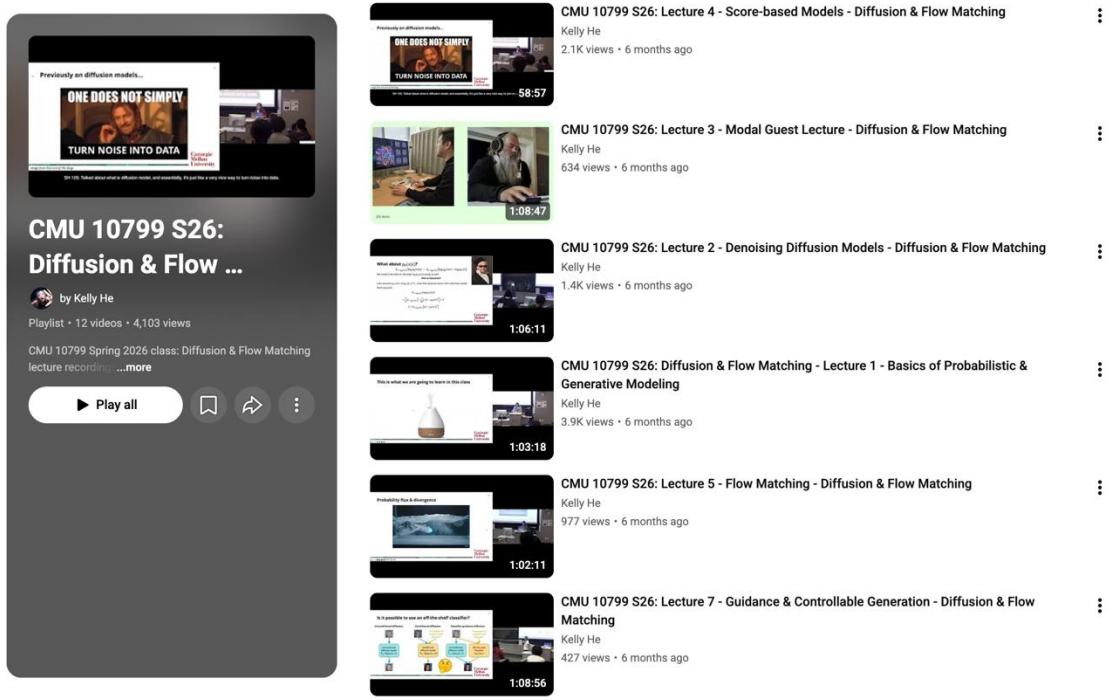
Plan for the Following Lectures

- **Diffusion and Flow-based Models**
  - ➢ Many fresh-new materials, featuring different introducing logistics



MIT 6.S184

Unified perspective of ODE / SDE



CMU 10.799 (S26)

Lectures on selected advanced topics



Lecture Outline

- **Introduction to Diffusion Models**
- **Learning Diffusion Models**
- **Diffusion Loss and Noise Parameterization**
- **Discussion**



Lecture Outline

- **Introduction to Diffusion Models**
- **Learning Diffusion Models**
- **Diffusion Loss and Noise Parameterization**
- **Discussion**

Introduction to Denoising Diffusion Model

- **The "Origin" of Diffusion**

Denoising Diffusion Probabilistic Models

Jonathan Ho
UC Berkeley
jonathanho@berkeley.edu

Ajay Jain
UC Berkeley
ajayj@berkeley.edu

Pieter Abbeel
UC Berkeley
pabbeel@cs.berkeley.edu

Abstract

We present high quality image synthesis results using diffusion probabilistic models, a class of latent variable models inspired by considerations from nonequilibrium thermodynamics. Our best results are obtained by training on a weighted variational bound designed according to a novel connection between diffusion probabilistic models and denoising score matching with Langevin dynamics, and our models naturally admit a progressive lossy decompression scheme that can be interpreted as a generalization of autoregressive decoding. On the unconditional CIFAR10 dataset, we obtain an Inception score of 9.46 and a state-of-the-art FID score of 3.17. On 256x256 LSUN, we obtain sample quality similar to ProgressiveGAN. Our implementation is available at https://github.com/hojonathanho/diffusion.

Ho et al., "Denoising Diffusion Probabilistic Models," **NeurIPS 2020**.

Introduction to Denoising Diffusion Model

- **The "True Origin" of Diffusion**

Deep Unsupervised Learning using Nonequilibrium Thermodynamics

Jascha Sohl-Dickstein JASCHA@STANFORD.EDU
Stanford University

Eric A. Weiss EAWEISS@BERKELEY.EDU
University of California, Berkeley

Niru Maheswaranathan NIRUM@STANFORD.EDU
Stanford University

Surya Ganguli SGANGULI@STANFORD.EDU
Stanford University

Sohl-Dickstein et al., "Deep Unsupervised Learning using Nonequilibrium Thermodynamics," **ICML 2015**.

Introduction to Denoising Diffusion Model

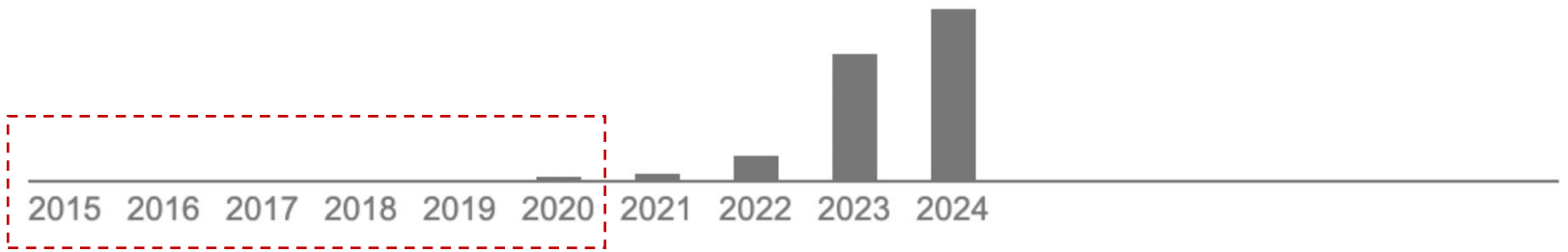
- **The "True Origin" of Diffusion**

Deep unsupervised learning using nonequilibrium thermodynamics

Authors	Jascha Sohl-Dickstein, Eric A Weiss, Niru Maheswaranathan, Surya Ganguli
Publication date	2015/3/12
Journal	International Conference on Machine Learning

Total citations: Cited by 5630 **Over 13,900 by Oct 2026!**

Largely Ignored



Sohl-Dickstein et al., "Deep Unsupervised Learning using Nonequilibrium Thermodynamics," **ICML 2015**.

Introduction to Denoising Diffusion Model

- **The Idea of Diffusion Process**

- **Forward Process:**

Destroy data structure

- **Reverse Process:**

Restore data structure



A Generative Model !

Abstract

A central problem in machine learning involves modeling complex data-sets using highly flexible families of probability distributions in which learning, sampling, inference, and evaluation are still analytically or computationally tractable. Here, we develop an approach that simultaneously achieves both flexibility and tractability. The essential idea, inspired by non-equilibrium statistical physics, is to systematically and slowly destroy structure in a data distribution through an iterative forward diffusion process. We then learn a reverse diffusion process that restores structure in data, yielding a highly flexible and tractable generative model of the data. This approach allows us to rapidly learn, sample from, and evaluate probabilities in deep generative models with thousands of layers or time steps, as well as to compute conditional and posterior probabilities under the learned model. We additionally release an open source reference implementation of the algorithm.

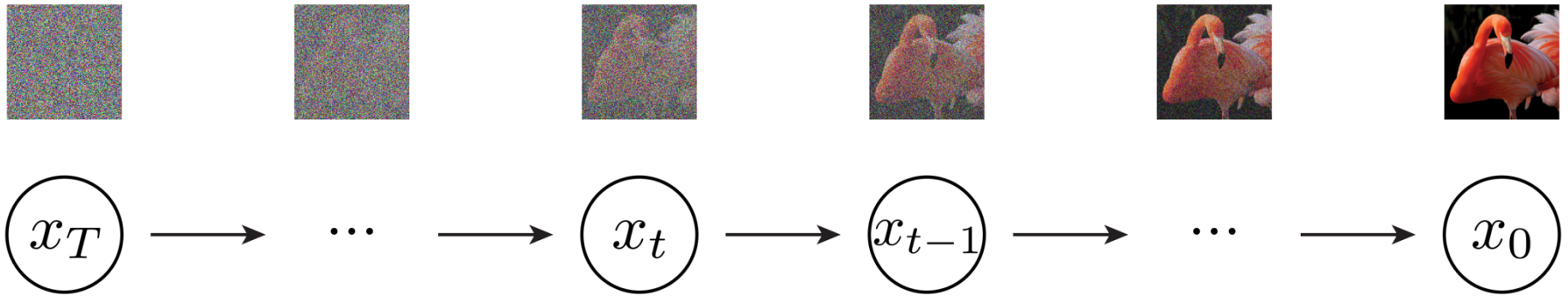
these models are unable to aptly describe structure in rich datasets. On the other hand, models that are *flexible* can be molded to fit structure in arbitrary data. For example, we can define models in terms of any (non-negative) function $\phi(\mathbf{x})$ yielding the flexible distribution $p(\mathbf{x}) = \frac{\phi(\mathbf{x})}{Z}$, where Z is a normalization constant. However, computing this normalization constant is generally intractable. Evaluating, training, or drawing samples from such flexible models typically requires a very expensive Monte Carlo process.

A variety of analytic approximations exist which ameliorate, but do not remove, this tradeoff–for instance mean field theory and its expansions (T, 1982; Tanaka, 1998), variational Bayes (Jordan et al., 1999), contrastive divergence (Welling & Hinton, 2002; Hinton, 2002), minimum probability flow (Sohl-Dickstein et al., 2011b;a), minimum KL contraction (Lyu, 2011), proper scoring rules (Gneiting & Raftery, 2007; Parry et al., 2012), score matching (Hyvärinen, 2005), pseudolikelihood (Besag, 1975), loopy belief propagation (Murphy et al., 1999), and many, many more. Non-parametric methods (Gershman & Blei, 2012) can also be very effective[1].

Sohl-Dickstein et al., "Deep Unsupervised Learning using Nonequilibrium Thermodynamics," ICML 2015.

Introduction to Denoising Diffusion Model

- **Diffusion Models in a Nutshell**
  - ➢ **Motivation:** Gradually turning data to noise, and learn the inverse process
    - ✓ **Forward** Process: **User-defined** noising process $q(x_t \mid x_{t-1})$
    - ✓ **Reverse** Process: **Learned** denoising process $p_\theta(x_{t-1} \mid x_t)$



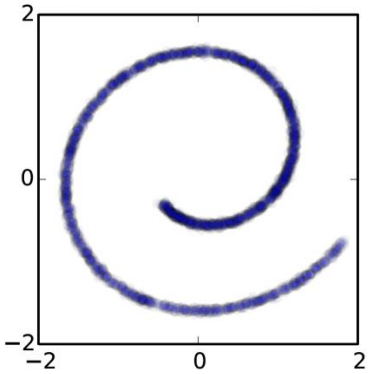
Introduction to Denoising Diffusion Model

- **Diffusion Models in a Nutshell**
  - **Motivation:** Gradually turning data to noise, and learn the inverse process

Forward Trajectory

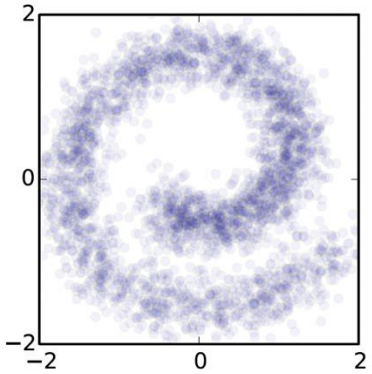
$q(x_{0:T})$

$t = 0$

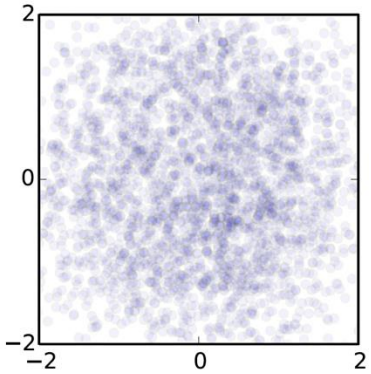


Clean Data Distribution

$t = \frac{T}{2}$



$t = T$

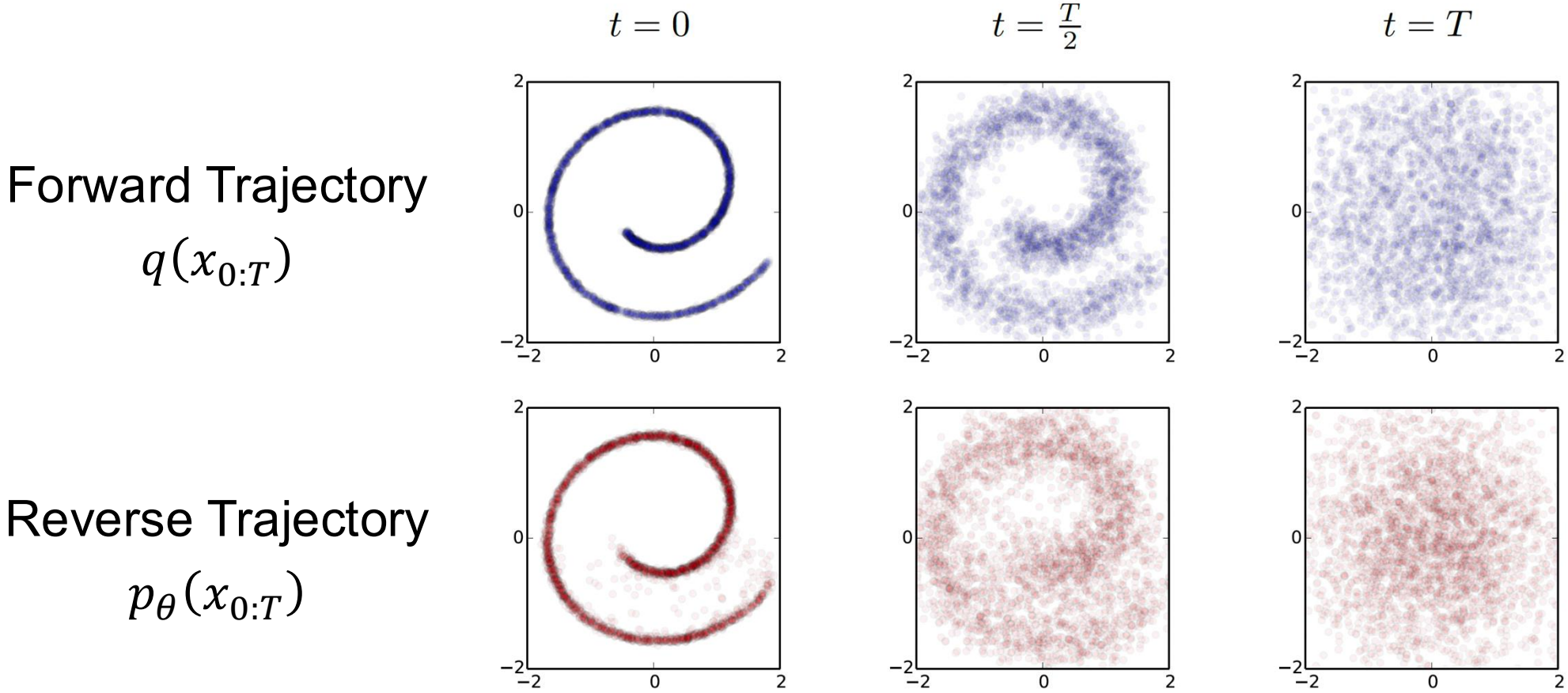


Pure Noise Distribution

Sohl-Dickstein et al., “Deep Unsupervised Learning using Nonequilibrium Thermodynamics,” ICML 2015.

Introduction to Denoising Diffusion Model

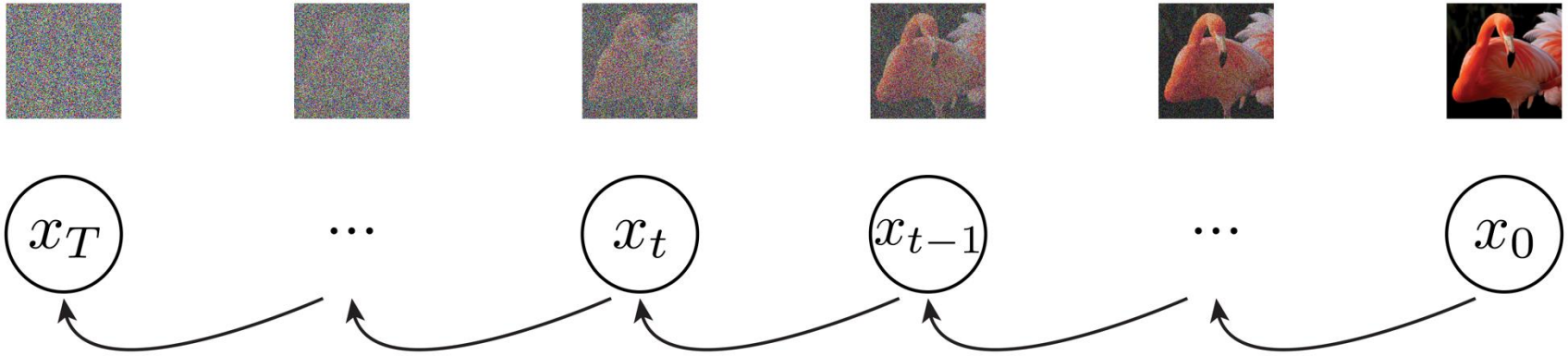
- **Diffusion Models in a Nutshell**
  - **Motivation:** Gradually turning data to noise, and learn the inverse process



Sohl-Dickstein et al., "Deep Unsupervised Learning using Nonequilibrium Thermodynamics," ICML 2015.

Introduction to Denoising Diffusion Model

- **Forward Process**
  - Data sample $x_0 \sim q(x_0)$
  - Run a **Markov Chain** that gradually adds noise to the data ➔ $x_1, \dots, x_T$



  - At each step t, we have

$$q(x_t \mid x_{t-1}) = \mathcal{N}(x_t; \sqrt{\alpha_t} x_{t-1}, (1-\alpha_t) I)$$

where $\alpha_t \in (0, 1)$. **Note:** $x_T \sim \mathcal{N}(0, I)$

- **Markov Property**
- **t: time step**
- **α_t: Schedule Param.**

Introduction to Denoising Diffusion Model

- **Forward Process**
  - **Note:** The distribution $q(x_1, \dots, x_T)$ has **analytically computable marginals**

$$q(x_t \mid x_{t-1}) = \mathcal{N}(x_t; \sqrt{\alpha_t} x_{t-1}, (1-\alpha_t) I)$$

Above is equivalent to $(\epsilon_t \sim \mathcal{N}(0, I))$

$$\begin{aligned}
\mathbf{x}_t &= \sqrt{\alpha_t}\mathbf{x}_{t-1} + \sqrt{1-\alpha_t}\boldsymbol{\epsilon}_t \\
&= \sqrt{\alpha_t\alpha_{t-1}}\mathbf{x}_{t-2} + \sqrt{1-\alpha_t\alpha_{t-1}}\bar{\boldsymbol{\epsilon}}_t \\
&= \dots \\
&= \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}
\end{aligned}$$

$$\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$$

$x_0 \to x_t$ in **closed-form** with **one step**

$$q(x_t \mid x_0) = \mathcal{N}\left(x_t; \sqrt{\bar{\alpha}_t} x_0, (1-\bar{\alpha}_t) I\right)$$

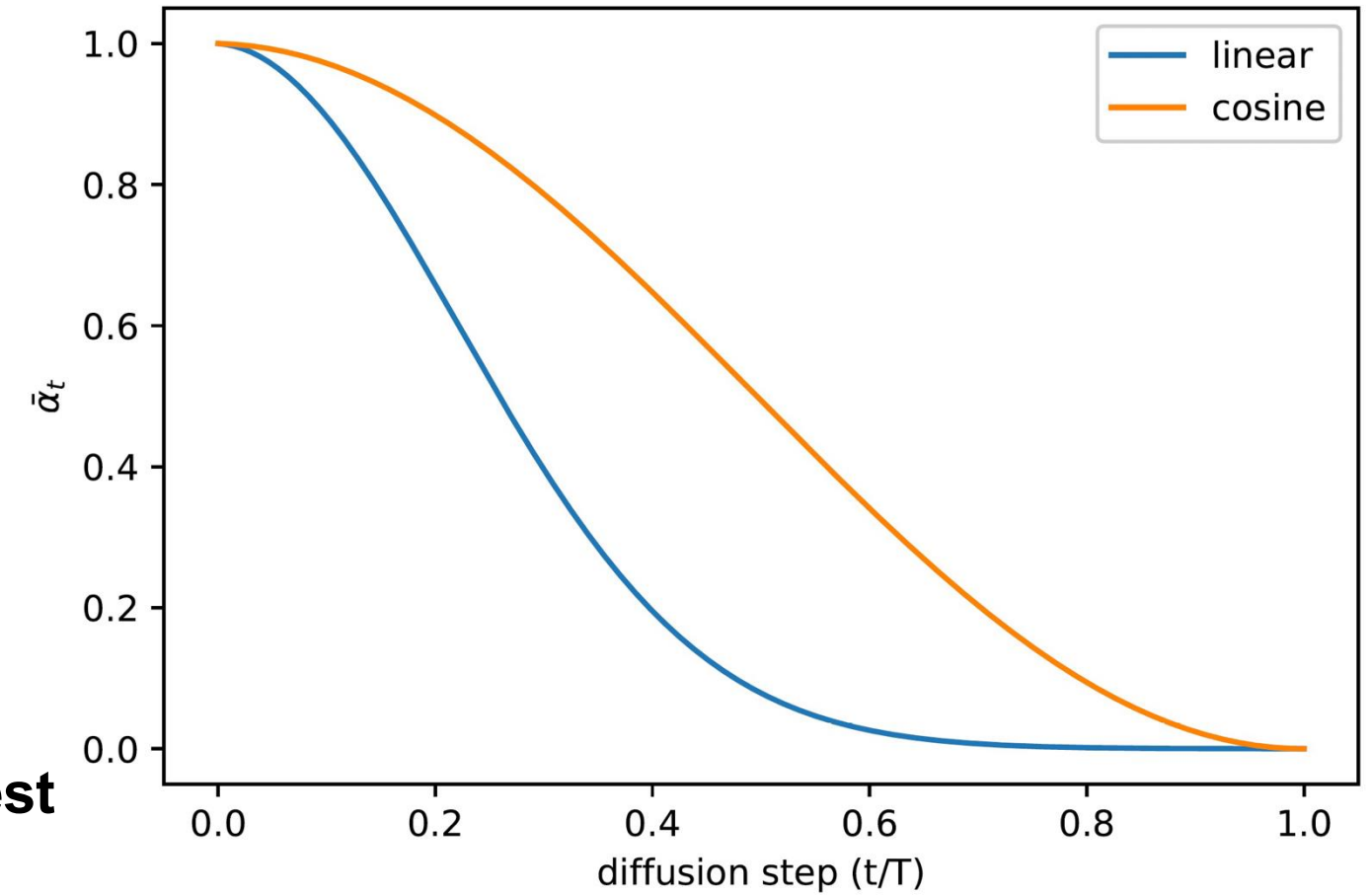
Sampling w/o Simulation

Introduction to Denoising Diffusion Model

- **Forward Process: Noise Scheduling**

$$x_t = \sqrt{\alpha_t} x_{t-1} + \sqrt{1-\alpha_t}\epsilon_t \quad \Rightarrow \quad x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon$$

Scheduling of $\bar{\alpha}_t$



Scheduling of α_t

Cleanest

Most Noisy

Introduction to Denoising Diffusion Model

- **Reverse Process**
  - **Problem:** The ideal denoising process $q(x_{t-1} \mid x_t)$ is **unknown**
    - ✓ Learn a model p_θ that **reverses** the Markovian noising process
  - **Solution:** Define p_θ with **latent variables** $x_1, \ldots, x_T$

$$p_\theta(x_{0:T}) = p(x_T) \prod_{t=1}^{T} p_\theta(x_{t-1} \mid x_t)$$

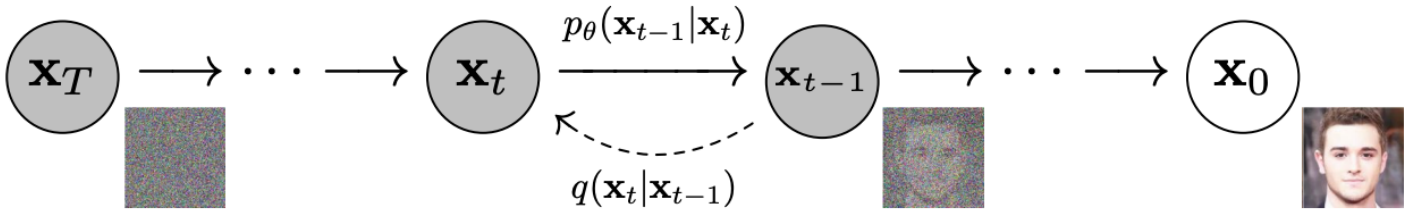
where the conditionals are

$$p_\theta(x_{t-1} \mid x_t) = \mathcal{N}\left(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t)\right)$$

➔ valid for **sufficiently small forward Gaussian noise steps** $(\alpha_t \to 1)$

  - **Sampling:** ① Sample noise x_T; ② Iteratively sample from $p_\theta(x_{t-1} \mid x_t)$

Introduction to Denoising Diffusion Model



- **Forward Process** (Pre-defined Markovian noising)

$$q(x_t \mid x_{t-1}) := \mathcal{N}(x_t; \sqrt{\alpha_t} x_{t-1}, (1-\alpha_t) I)$$

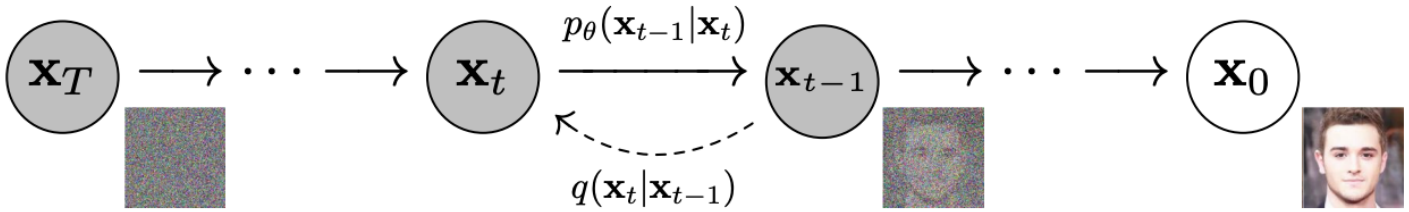
$$q(x_{1:T} \mid x_0) := \prod_{t=1}^{T} q(x_t \mid x_{t-1})$$

$$\begin{cases} x_t(x_{t-1}) = \sqrt{\alpha_t} x_{t-1} + \sqrt{1-\alpha_t}\epsilon_t \\ x_t(x_0) = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon \end{cases}$$

- **Reverse Process** (Learned de-noising model with parameter θ)

$$p_\theta(x_{0:T}) := p(x_T) \prod_{t=1}^{T} p_\theta(x_{t-1} \mid x_t) \qquad p_\theta(x_{t-1} \mid x_t) := \mathcal{N}\left(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t)\right)$$

Introduction to Denoising Diffusion Model



- **Forward Process** (Pre-defined Markovian noising)

$$\alpha_t = 1 - \beta_t$$

$$q(x_t \mid x_{t-1}) := \mathcal{N}(x_t; \sqrt{\alpha_t} x_{t-1}, \beta_t I)$$

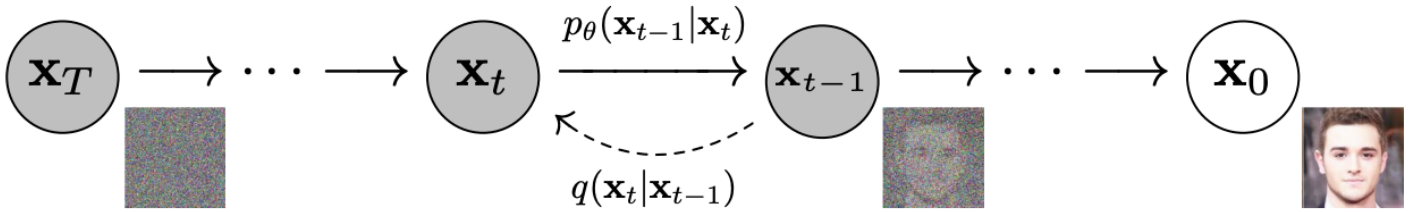
$$q(x_{1:T} \mid x_0) := \prod_{t=1}^{T} q(x_t \mid x_{t-1})$$

$$\begin{cases} x_t(x_{t-1}) = \sqrt{\alpha_t} x_{t-1} + \sqrt{\beta_t}\epsilon_t \\ x_t(x_0) = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon \end{cases}$$

- **Reverse Process** (Learned de-noising model with parameter θ)

$$p_\theta(x_{0:T}) := p(x_T) \prod_{t=1}^{T} p_\theta(x_{t-1} \mid x_t) \qquad p_\theta(x_{t-1} \mid x_t) := \mathcal{N}\left(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t)\right)$$

Introduction to Denoising Diffusion Model



- **Noise Scheduling Parameters**

$$\alpha_t = 1 - \beta_t \quad \Rightarrow \quad \begin{cases} x_t(x_{t-1}) = \sqrt{\alpha_t}x_{t-1} + \sqrt{\beta_t}\epsilon_t \\ \bar{\alpha}_t = \alpha_t\bar{\alpha}_{t-1}, \quad \alpha_t = \dfrac{\bar{\alpha}_t}{\bar{\alpha}_{t-1}}, \quad \beta_t = 1 - \dfrac{\bar{\alpha}_t}{\bar{\alpha}_{t-1}} \end{cases}$$

$$\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s = \prod_{s=1}^{t}(1-\beta_s) \quad \Rightarrow \quad \bar{\beta}_t := 1 - \bar{\alpha}_t = 1 - \prod_{s=1}^{t}(1-\beta_s)$$



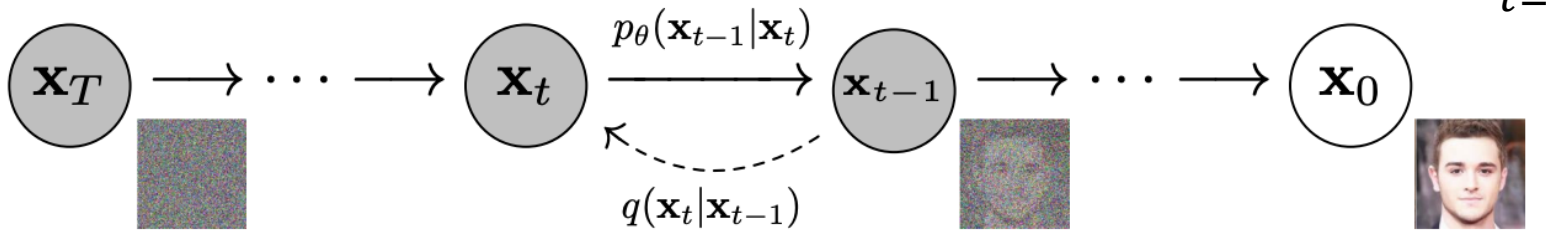
Lecture Outline

- Introduction to Diffusion Models
- Learning Diffusion Models
- Diffusion Loss and Noise Parameterization
- Discussion

Learning Diffusion Models

- **Learning Reverse from Forward**

$$p_\theta(x_{0:T}) := p(x_T) \prod_{t=1}^{T} p_\theta(x_{t-1} \mid x_t)$$



➢ **Goal:** Learn $p_\theta(x_{t-1} \mid x_t)$ via **Maximum Likelihood**

$$\log p_\theta(x_0) = \log \int p_\theta(x_{0:T})\, dx_{1:T}$$

$$= \log \int q(x_{1:T} \mid x_0) \frac{p_\theta(x_{0:T})}{q(x_{1:T} \mid x_0)}\, dx_{1:T}$$

Jensen's Inequality

$$= \log \mathbb{E}_{q(x_{1:T}|x_0)} \left[\frac{p_\theta(x_{0:T})}{q(x_{1:T} \mid x_0)} \right] \geq \mathbb{E}_{q(x_{1:T}|x_0)} \left[\log \frac{p_\theta(x_{0:T})}{q(x_{1:T} \mid x_0)} \right]$$

ELBO

Sohl-Dickstein et al., "Deep Unsupervised Learning using Nonequilibrium Thermodynamics," ICML 2015.

Learning Diffusion Models

- **ELBO Derivation from Another Perspective**
  - **Goal: Minimizing the KL Divergence** between the two processes

$$D_{KL}\big(q(x_{1:T} \mid x_0) \,\|\, p_\theta(x_{1:T} \mid x_0)\big)$$

  - ➔ learn $p_\theta(x_{1:T} \mid x_0)$ that matches the forward noising trajectory induced by x_0

$$\begin{aligned}
\log p_\theta(\mathbf{x}_0) &\geq \log p_\theta(\mathbf{x}_0) - D_{\mathrm{KL}}(q(\mathbf{x}_{1:T}|\mathbf{x}_0)\|p_\theta(\mathbf{x}_{1:T}|\mathbf{x}_0)) \\
&= \log p_\theta(\mathbf{x}_0) - \mathbb{E}_{\mathbf{x}_{1:T}\sim q(\mathbf{x}_{1:T}|\mathbf{x}_0)}\left[\log \frac{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}{p_\theta(\mathbf{x}_{0:T})/p_\theta(\mathbf{x}_0)}\right] \\
&= \log p_\theta(\mathbf{x}_0) - \mathbb{E}_q\left[\log \frac{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}{p_\theta(\mathbf{x}_{0:T})} + \log p_\theta(\mathbf{x}_0)\right] \\
&= -\mathbb{E}_q\left[\log \frac{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}{p_\theta(\mathbf{x}_{0:T})}\right]
\end{aligned}$$

Similar to VAE's objective, but q is fixed and hand-crafted

$$\mathbb{E}_{\mathbf{z}\sim q(\mathbf{z})}\left[\frac{p_\theta(\mathbf{x},\mathbf{z})}{q(\mathbf{z})}\right]$$

Learning Diffusion Models

- **From ELBO to the Objective Function**

$$L = \mathbb{E}_q \left[-\log \frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \right]$$

Starting from the ELBO of Diffusion Model

Ho et al., “Denoising Diffusion Probabilistic Models,” NeurIPS 2020.

Learning Diffusion Models

- **From ELBO to the Objective Function**

$$L = \mathbb{E}_q\left[-\log \frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}\right]$$

$$= \mathbb{E}_q\left[-\log p(\mathbf{x}_T) - \sum_{t\geq 1} \log \frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_t|\mathbf{x}_{t-1})}\right]$$

$$q(x_{1:T} \mid x_0) = \prod_{t=1}^{T} q(x_t \mid x_{t-1})$$

$$p_\theta(x_{0:T}) = p(x_T) \prod_{t=1}^{T} p_\theta(x_{t-1} \mid x_t)$$

(Markov Property)

Ho et al., "Denoising Diffusion Probabilistic Models," NeurIPS 2020.

Learning Diffusion Models

- **From ELBO to the Objective Function**

$$
\begin{aligned}
L &= \mathbb{E}_q \left[-\log \frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \right] \\
&= \mathbb{E}_q \left[-\log p(\mathbf{x}_T) - \sum_{t \geq 1} \log \frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_t|\mathbf{x}_{t-1})} \right] \\
&= \mathbb{E}_q \left[-\log p(\mathbf{x}_T) - \sum_{t > 1} \log \frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_t|\mathbf{x}_{t-1})} - \log \frac{p_\theta(\mathbf{x}_0|\mathbf{x}_1)}{q(\mathbf{x}_1|\mathbf{x}_0)} \right]
\end{aligned}
$$



Leave $t = 1$ term out for separate processing

Ho et al., "Denoising Diffusion Probabilistic Models," NeurIPS 2020.

Learning Diffusion Models

- **From ELBO to the Objective Function**

$$
\begin{aligned}
L &= \mathbb{E}_q \left[-\log \frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \right] \\
&= \mathbb{E}_q \left[-\log p(\mathbf{x}_T) - \sum_{t \geq 1} \log \frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_t|\mathbf{x}_{t-1})} \right] \\
&= \mathbb{E}_q \left[-\log p(\mathbf{x}_T) - \sum_{t > 1} \log \frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_t|\mathbf{x}_{t-1})} - \log \frac{p_\theta(\mathbf{x}_0|\mathbf{x}_1)}{q(\mathbf{x}_1|\mathbf{x}_0)} \right] \\
&= \mathbb{E}_q \left[-\log p(\mathbf{x}_T) - \sum_{t > 1} \log \frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)} \cdot \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_0)}{q(\mathbf{x}_t|\mathbf{x}_0)} - \log \frac{p_\theta(\mathbf{x}_0|\mathbf{x}_1)}{q(\mathbf{x}_1|\mathbf{x}_0)} \right]
\end{aligned}
$$

Bayes' Equation

Ho et al., "Denoising Diffusion Probabilistic Models," NeurIPS 2020.

Learning Diffusion Models

- **From ELBO to the Objective Function**

$$
\begin{aligned}
L &= \mathbb{E}_q\left[-\log \frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}\right] \\
&= \mathbb{E}_q\left[-\log p(\mathbf{x}_T) - \sum_{t\geq 1} \log \frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_t|\mathbf{x}_{t-1})}\right] \\
&= \mathbb{E}_q\left[-\log p(\mathbf{x}_T) - \sum_{t>1} \log \frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_t|\mathbf{x}_{t-1})} - \log \frac{p_\theta(\mathbf{x}_0|\mathbf{x}_1)}{q(\mathbf{x}_1|\mathbf{x}_0)}\right] \\
&= \mathbb{E}_q\left[-\log p(\mathbf{x}_T) - \sum_{t>1} \log \frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)} \cdot \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_0)}{q(\mathbf{x}_t|\mathbf{x}_0)} - \log \frac{p_\theta(\mathbf{x}_0|\mathbf{x}_1)}{q(\mathbf{x}_1|\mathbf{x}_0)}\right]
\end{aligned}
$$

Bayes' Equation

$$
q(x_{t-1} \mid x_t, x_0) = \frac{q(x_t \mid x_{t-1}, x_0)\, q(x_{t-1} \mid x_0)}{q(x_t \mid x_0)} \Rightarrow q(x_{t-1} \mid x_t, x_0) = \frac{q(x_t \mid x_{t-1})\, q(x_{t-1} \mid x_0)}{q(x_t \mid x_0)}
$$

Markov property

Ho et al., "Denoising Diffusion Probabilistic Models," NeurIPS 2020.

Learning Diffusion Models

- **From ELBO to the Objective Function**

$$
\begin{aligned}
L &= \mathbb{E}_q \left[-\log \frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \right] \\
&= \mathbb{E}_q \left[-\log p(\mathbf{x}_T) - \sum_{t \geq 1} \log \frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_t|\mathbf{x}_{t-1})} \right] \\
&= \mathbb{E}_q \left[-\log p(\mathbf{x}_T) - \sum_{t > 1} \log \frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_t|\mathbf{x}_{t-1})} - \log \frac{p_\theta(\mathbf{x}_0|\mathbf{x}_1)}{q(\mathbf{x}_1|\mathbf{x}_0)} \right] \\
&= \mathbb{E}_q \left[-\log p(\mathbf{x}_T) - \sum_{t > 1} \log \frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)} \cdot \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_0)}{q(\mathbf{x}_t|\mathbf{x}_0)} - \log \frac{p_\theta(\mathbf{x}_0|\mathbf{x}_1)}{q(\mathbf{x}_1|\mathbf{x}_0)} \right]
\end{aligned}
$$

Bayes' Equation

$$
q(x_{t-1} \mid x_t, x_0) = \frac{q(x_t \mid x_{t-1})\, q(x_{t-1} \mid x_0)}{q(x_t \mid x_0)} \quad \Rightarrow \quad q(x_t \mid x_{t-1}) = \frac{q(x_{t-1} \mid x_t, x_0)\, q(x_t \mid x_0)}{q(x_{t-1} \mid x_0)}
$$

Ho et al., "Denoising Diffusion Probabilistic Models," NeurIPS 2020.

Learning Diffusion Models

- **From ELBO to the Objective Function**

$$
\begin{aligned}
L &= \mathbb{E}_q\left[-\log \frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}\right] \\
&= \mathbb{E}_q\left[-\log p(\mathbf{x}_T) - \sum_{t\geq 1} \log \frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_t|\mathbf{x}_{t-1})}\right] \\
&= \mathbb{E}_q\left[-\log p(\mathbf{x}_T) - \sum_{t>1} \log \frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_t|\mathbf{x}_{t-1})} - \log \frac{p_\theta(\mathbf{x}_0|\mathbf{x}_1)}{q(\mathbf{x}_1|\mathbf{x}_0)}\right] \\
&= \mathbb{E}_q\left[-\log p(\mathbf{x}_T) - \sum_{t>1} \log \frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)} \cdot \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_0)}{q(\mathbf{x}_t|\mathbf{x}_0)} - \log \frac{p_\theta(\mathbf{x}_0|\mathbf{x}_1)}{q(\mathbf{x}_1|\mathbf{x}_0)}\right]
\end{aligned}
$$

Bayes' Equation

$$
q(x_{t-1} \mid x_t, x_0) = \frac{q(x_t \mid x_{t-1})\, q(x_{t-1} \mid x_0)}{q(x_t \mid x_0)} \quad \Rightarrow \quad q(x_t \mid x_{t-1}) = \frac{q(x_{t-1} \mid x_t, x_0)\, q(x_t \mid x_0)}{q(x_{t-1} \mid x_0)}
$$

Why would we do this?

Ho et al., "Denoising Diffusion Probabilistic Models," NeurIPS 2020.

Learning Diffusion Models

- **From ELBO to the Objective Function**

$$
\begin{aligned}
L &= \mathbb{E}_q\left[-\log \frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}\right] \\
&= \mathbb{E}_q\left[-\log p(\mathbf{x}_T) - \sum_{t\geq 1} \log \frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_t|\mathbf{x}_{t-1})}\right] \\
&= \mathbb{E}_q\left[-\log p(\mathbf{x}_T) - \sum_{t>1} \log \frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_t|\mathbf{x}_{t-1})} - \log \frac{p_\theta(\mathbf{x}_0|\mathbf{x}_1)}{q(\mathbf{x}_1|\mathbf{x}_0)}\right] \\
&= \mathbb{E}_q\left[-\log p(\mathbf{x}_T) - \sum_{t>1} \log \frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)} \cdot \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_0)}{q(\mathbf{x}_t|\mathbf{x}_0)} - \log \frac{p_\theta(\mathbf{x}_0|\mathbf{x}_1)}{q(\mathbf{x}_1|\mathbf{x}_0)}\right]
\end{aligned}
$$

Mis-aligned Variables

$(x_{t-1} \mid x_t)$ vs. $(x_t \mid x_{t-1})$

Bayes' Equation

Learning Diffusion Models

- **From ELBO to the Objective Function**

$$L = \mathbb{E}_q\left[-\log \frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}\right]$$

$$= \mathbb{E}_q\left[-\log p(\mathbf{x}_T) - \sum_{t\geq 1} \log \frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_t|\mathbf{x}_{t-1})}\right]$$

$$= \mathbb{E}_q\left[-\log p(\mathbf{x}_T) - \sum_{t>1} \log \frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_t|\mathbf{x}_{t-1})} - \log \frac{p_\theta(\mathbf{x}_0|\mathbf{x}_1)}{q(\mathbf{x}_1|\mathbf{x}_0)}\right]$$

$$= \mathbb{E}_q\left[-\log p(\mathbf{x}_T) - \sum_{t>1} \log \frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)} \cdot \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_0)}{q(\mathbf{x}_t|\mathbf{x}_0)} - \log \frac{p_\theta(\mathbf{x}_0|\mathbf{x}_1)}{q(\mathbf{x}_1|\mathbf{x}_0)}\right]$$

Mis-aligned Variables

$(x_{t-1} \mid x_t)$ vs. $(x_t \mid x_{t-1})$

Bayes' Equation

Aligned Variables

$(x_{t-1} \mid x_t)$ vs. $(x_{t-1} \mid x_t, x_0)$

KL Divergence can be computed

Learning Diffusion Models

- **From ELBO to the Objective Function**

$$L = \mathbb{E}_q\left[-\log \frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}\right]$$

$$= \mathbb{E}_q\left[-\log p(\mathbf{x}_T) - \sum_{t\geq 1} \log \frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_t|\mathbf{x}_{t-1})}\right]$$

$$= \mathbb{E}_q\left[-\log p(\mathbf{x}_T) - \sum_{t>1} \log \frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_t|\mathbf{x}_{t-1})} - \log \frac{p_\theta(\mathbf{x}_0|\mathbf{x}_1)}{q(\mathbf{x}_1|\mathbf{x}_0)}\right]$$

$$= \mathbb{E}_q\left[-\log p(\mathbf{x}_T) - \sum_{t>1} \log \frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)} \cdot \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_0)}{q(\mathbf{x}_t|\mathbf{x}_0)} - \log \frac{p_\theta(\mathbf{x}_0|\mathbf{x}_1)}{q(\mathbf{x}_1|\mathbf{x}_0)}\right]$$

$$\sum_{t=2}^{T} \log \frac{q(x_{t-1} \mid x_0)}{q(x_t \mid x_0)} = \log\left[\prod_{t=2}^{T} \frac{q(x_{t-1} \mid x_0)}{q(x_t \mid x_0)}\right]$$

$$= \log\left[\frac{q(x_1 \mid x_0)}{q(x_2 \mid x_0)} \frac{q(x_2 \mid x_0)}{q(x_3 \mid x_0)} \cdots \frac{q(x_{T-1} \mid x_0)}{q(x_T \mid x_0)}\right]$$

$$= \log \frac{q(x_1 \mid x_0)}{q(x_T \mid x_0)}$$

Telescoping Cancellation
(逐项消元)

Ho et al., "Denoising Diffusion Probabilistic Models," NeurIPS 2020.

Learning Diffusion Models

- **From ELBO to the Objective Function**

$$L = \mathbb{E}_q\left[-\log \frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}\right]$$

$$= \mathbb{E}_q\left[-\log p(\mathbf{x}_T) - \sum_{t\geq 1} \log \frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_t|\mathbf{x}_{t-1})}\right]$$

$$= \mathbb{E}_q\left[-\log p(\mathbf{x}_T) - \sum_{t>1} \log \frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_t|\mathbf{x}_{t-1})} - \log \frac{p_\theta(\mathbf{x}_0|\mathbf{x}_1)}{q(\mathbf{x}_1|\mathbf{x}_0)}\right]$$

$$= \mathbb{E}_q\left[-\log p(\mathbf{x}_T) - \sum_{t>1} \log \frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)} \cdot \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_0)}{q(\mathbf{x}_t|\mathbf{x}_0)} - \log \frac{p_\theta(\mathbf{x}_0|\mathbf{x}_1)}{q(\mathbf{x}_1|\mathbf{x}_0)}\right]$$

$$= \mathbb{E}_q\left[-\log \frac{p(\mathbf{x}_T)}{q(\mathbf{x}_T|\mathbf{x}_0)} - \sum_{t>1} \log \frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)} - \log p_\theta(\mathbf{x}_0|\mathbf{x}_1)\right]$$

$$\sum_{t=2}^{T} \log \frac{q(x_{t-1} \mid x_0)}{q(x_t \mid x_0)} = \log\left[\prod_{t=2}^{T} \frac{q(x_{t-1} \mid x_0)}{q(x_t \mid x_0)}\right]$$

$$= \log\left[\frac{q(x_1 \mid x_0)}{q(x_2 \mid x_0)} \frac{q(x_2 \mid x_0)}{q(x_3 \mid x_0)} \cdots \frac{q(x_{T-1} \mid x_0)}{q(x_T \mid x_0)}\right]$$

$$= \log \frac{q(x_1 \mid x_0)}{q(x_T \mid x_0)}$$

Simple Algebra

Ho et al., "Denoising Diffusion Probabilistic Models," NeurIPS 2020.

Learning Diffusion Models

- **From ELBO to the Objective Function**

$$L = \mathbb{E}_q\left[-\log \frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}\right]$$

$$= \mathbb{E}_q\left[-\log \frac{p(\mathbf{x}_T)}{q(\mathbf{x}_T|\mathbf{x}_0)} - \sum_{t>1} \log \frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)} - \log p_\theta(\mathbf{x}_0|\mathbf{x}_1)\right]$$

$$= \mathbb{E}_q\left[\underbrace{D_{\mathrm{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \parallel p(\mathbf{x}_T))}_{L_T} + \sum_{t>1} \underbrace{D_{\mathrm{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))}_{L_t} - \underbrace{\log p_\theta(\mathbf{x}_0|\mathbf{x}_1)}_{L_0}\right]$$

Prior Matching L_T

Trajectory Matching L_t

Reconstruction Matching L_0

Ho et al., "Denoising Diffusion Probabilistic Models," NeurIPS 2020.

Learning Diffusion Models

$$\mathbb{E}_q \left[D_{\mathrm{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \parallel p(\mathbf{x}_T)) + \sum_{t>1} D_{\mathrm{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)) - \log p_\theta(\mathbf{x}_0|\mathbf{x}_1) \right]$$

- **Prior Loss:** $L_T = D_{\mathrm{KL}}\big(q(x_T \mid x_0) \,\|\, p(x_T)\big)$
  - ✓ Compares the distribution of noisy data x_T in two processes
  - ➔ **Can often reduce to zero by construction**

$$x_T(x_0) = \sqrt{\bar{\alpha}_T} x_0 + \sqrt{1 - \bar{\alpha}_T}\epsilon$$

$$\bar{\alpha}_T = \prod_{s=1}^{T} \alpha_s$$

Learning Diffusion Models

$$\mathbb{E}_q \left[D_{\mathrm{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \parallel p(\mathbf{x}_T)) + \sum_{t>1} D_{\mathrm{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)) - \log p_\theta(\mathbf{x}_0|\mathbf{x}_1) \right]$$

- **Prior Loss:** $L_T = D_{\mathrm{KL}}\big(q(x_T \mid x_0) \,\|\, p(x_T)\big)$
  - ✓ Compares noisy data x_T ➔ Can reduce to zero by construction
- **Reconstruction Loss:** $L_0 = -\log p_\theta\,(x_0 \mid x_1)$
  - ✓ Recover the clean data x_0 given the closest noisy latent state x_1
  - ✓ Similar to the reconstruction loss in VAE

Ho et al., "Denoising Diffusion Probabilistic Models," NeurIPS 2020.

Learning Diffusion Models

$$\mathbb{E}_q \left[D_{\mathrm{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \parallel p(\mathbf{x}_T)) + \sum_{t>1} D_{\mathrm{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)) - \log p_\theta(\mathbf{x}_0|\mathbf{x}_1) \right]$$

- **Prior Loss:** $L_T = D_{\mathrm{KL}}\big(q(x_T \mid x_0) \,\|\, p(x_T)\big)$
  - ✓ Compares noisy data x_T ➔ Can reduce to zero by construction
- **Reconstruction Loss:** $L_0 = -\log p_\theta\,(x_0 \mid x_1)$
  - ✓ Recover the clean data x_0 given the closest noisy latent state x_1
- **Diffusion Loss** **Why not using $q(x_{t-1} \mid x_t)$?**

$$L_t = D_{KL}\big(q(x_{t-1} \mid x_t, x_0) \,\|\, p_\theta(x_{t-1} \mid x_t)\big)$$

  - ✓ For $1 < t \leq T$, measures how well the **learned reverse transition** $p_\theta(x_{t-1} \mid x_t)$ matches the **forward posterior** $q(x_{t-1} \mid x_t, x_0)$

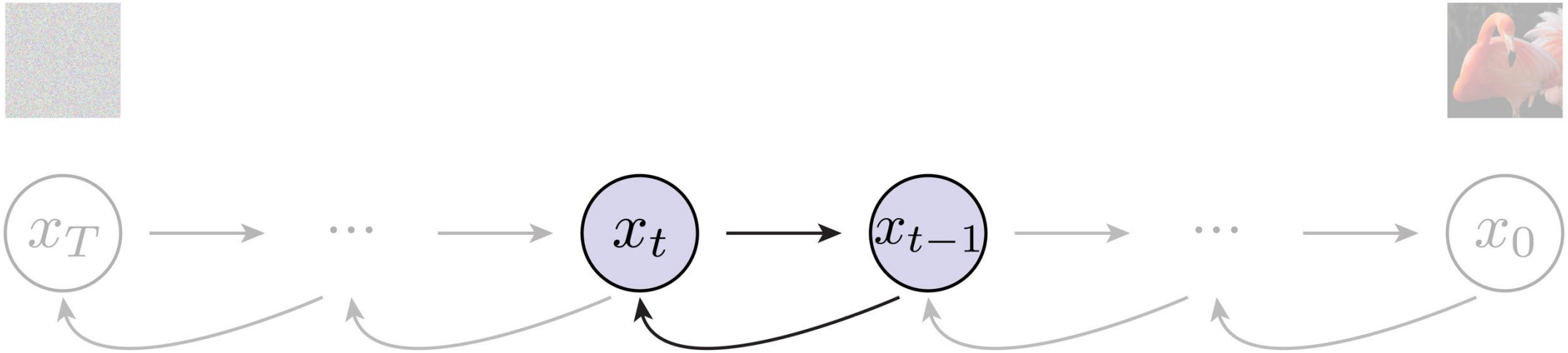


Lecture Outline

- Introduction to Diffusion Models
- Learning Diffusion Models
- Diffusion Loss and Noise Parameterization
- Discussion

Diffusion Loss and Noise Parameterization

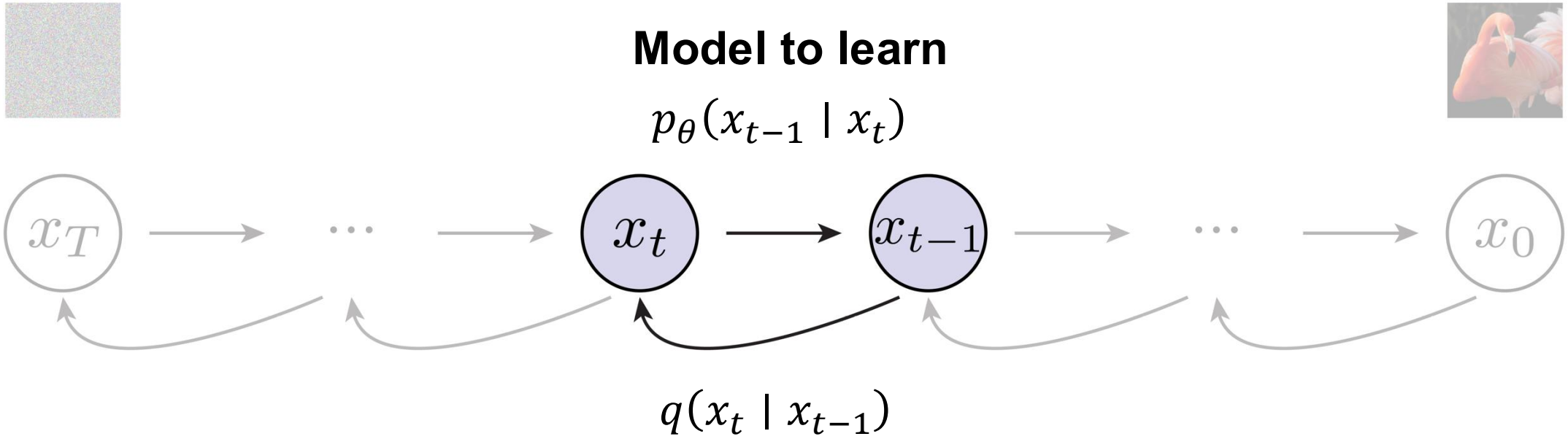
- **Forward Process**



Diffusion Model with Gaussian Noise

Diffusion Loss and Noise Parameterization

- **Forward Process**

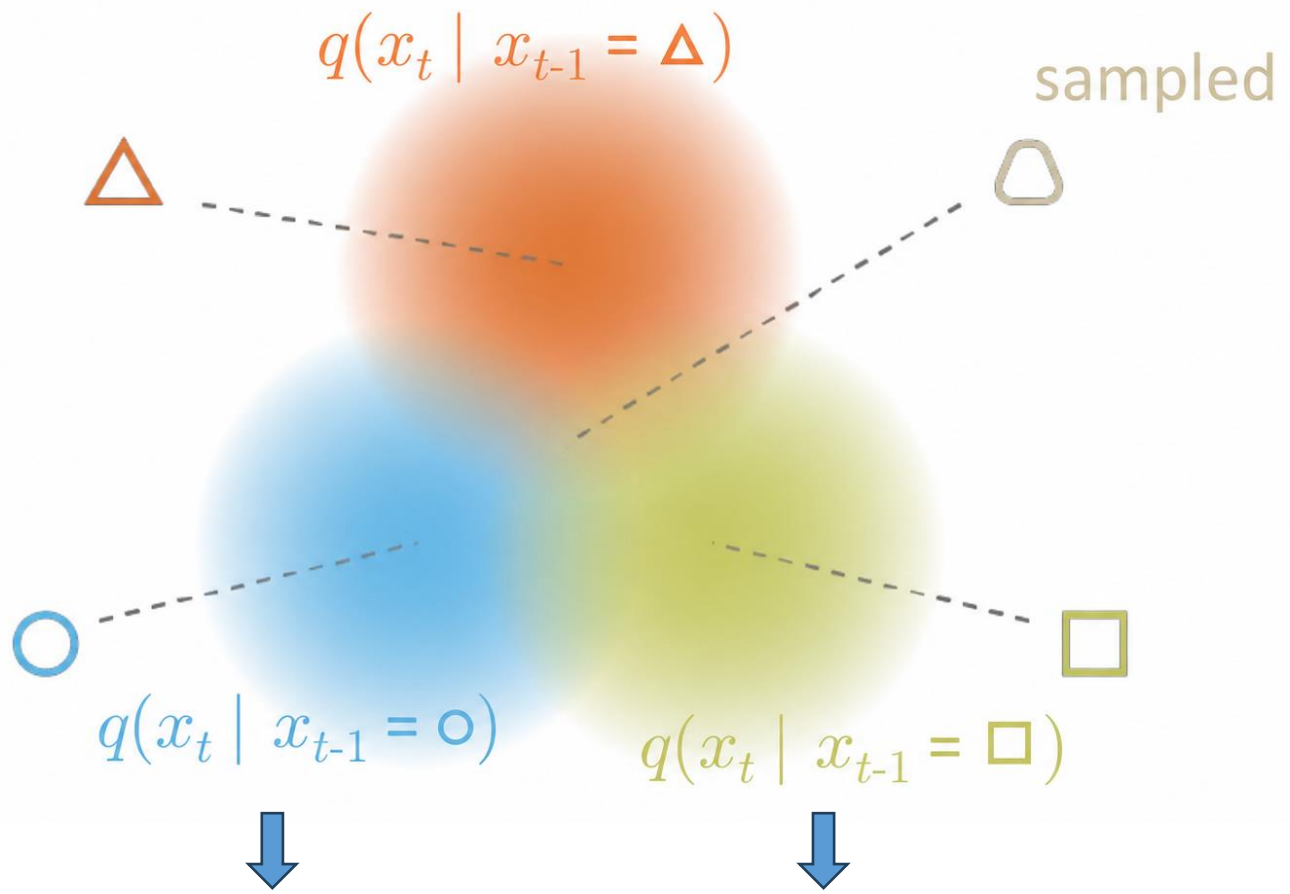


Known process, but

not our target

Diffusion Loss and Noise Parameterization

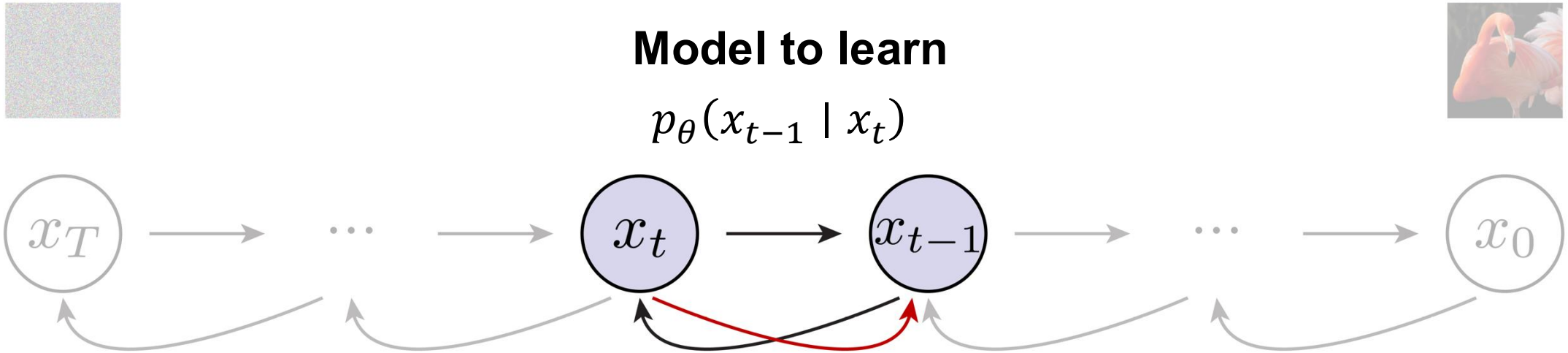
- **Visualization: Forward Process**



Forward conditionals are known (Gaussian)

Diffusion Loss and Noise Parameterization

- **Reverse Process**

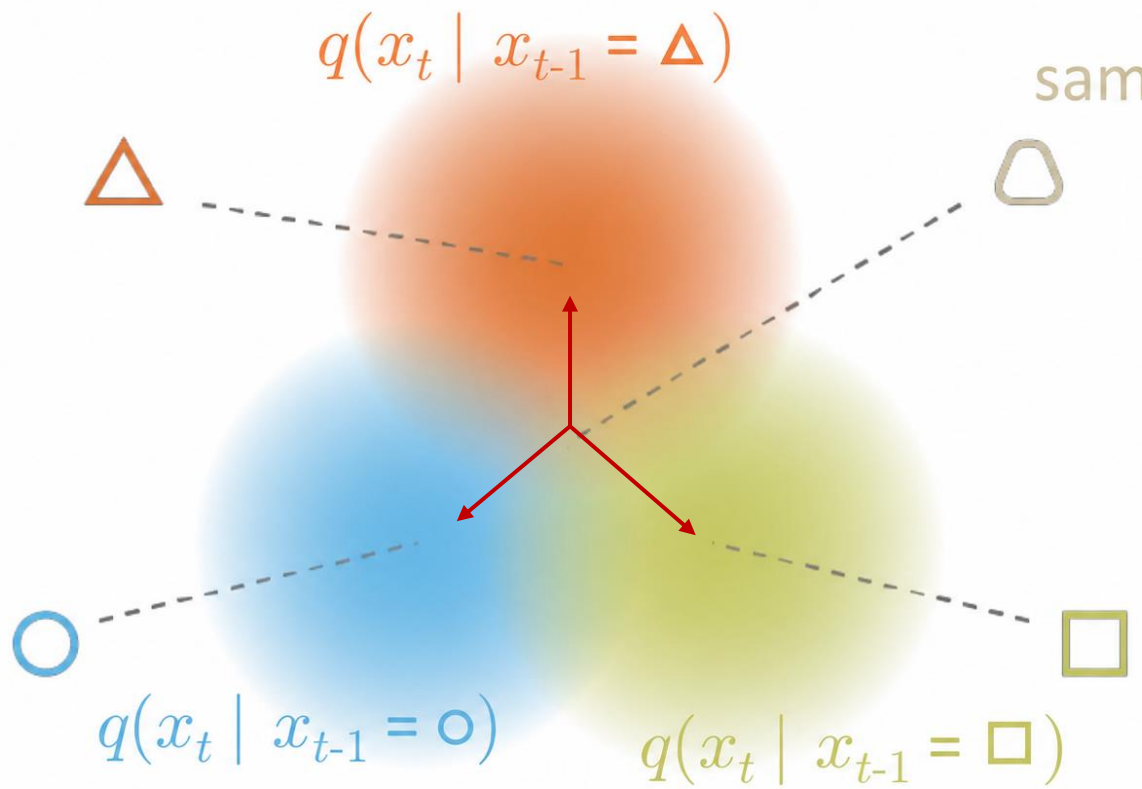


Our target,

but **unknown!**

Diffusion Loss and Noise Parameterization

- **Visualization: Reverse Process**



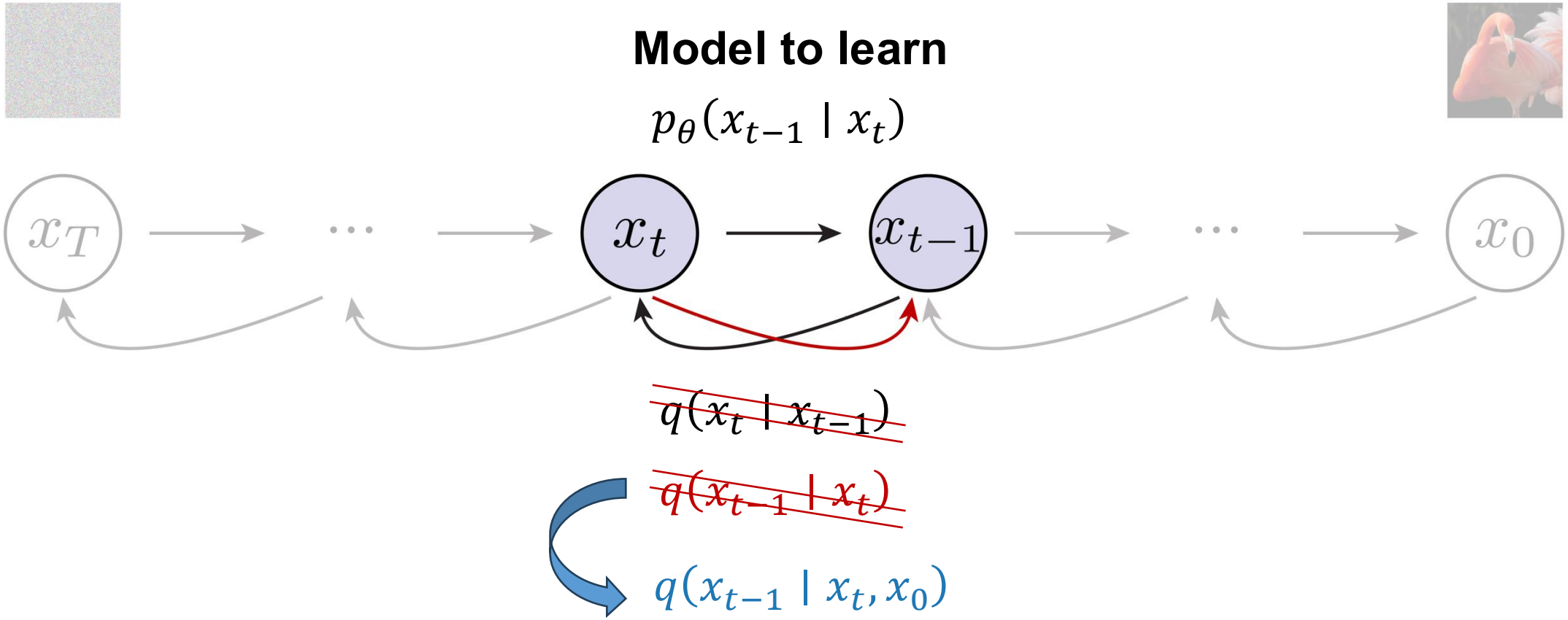
Reverse conditionals are unknown

$q(x_{t-1} \mid x_t =$  $)$

Need to learn from the dataset

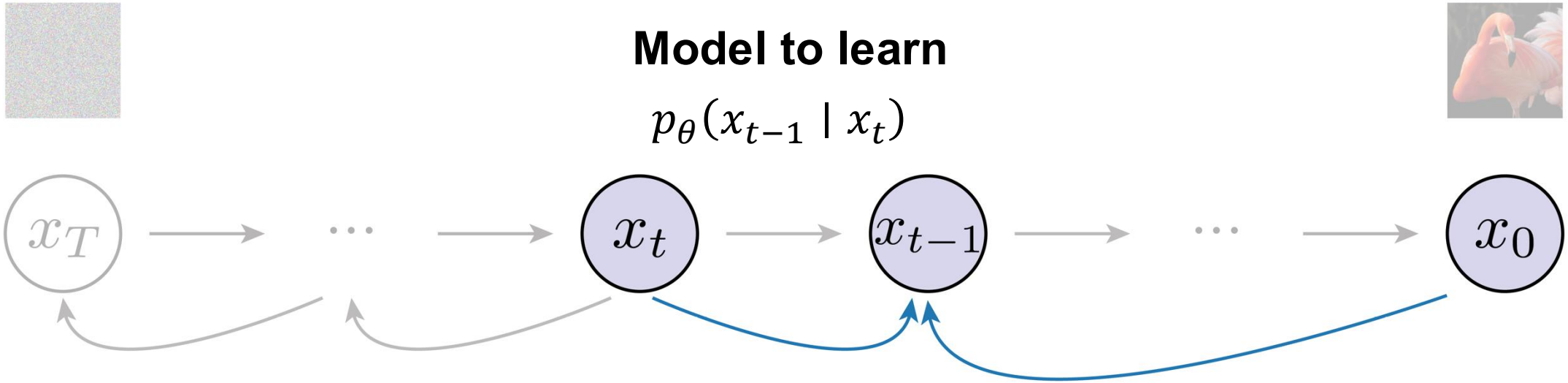
Diffusion Loss and Noise Parameterization

- **Posterior Conditioned on x_0**



Diffusion Loss and Noise Parameterization

- **Posterior Conditioned on x_0**

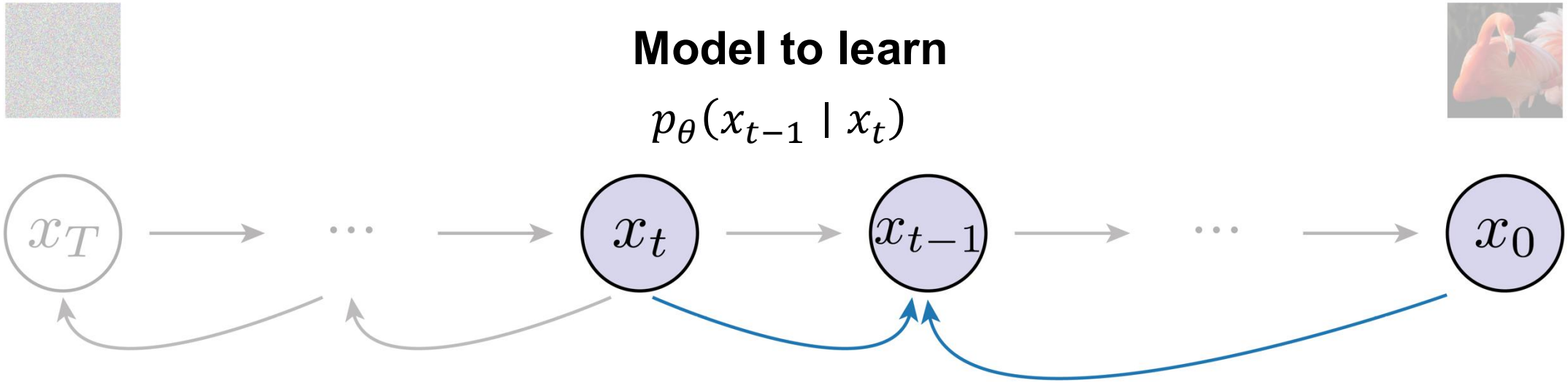


$$q(x_{t-1} \mid x_t, x_0) = \frac{q(x_t, x_{t-1} \mid x_0)}{q(x_t \mid x_0)} = \frac{q(x_t \mid x_{t-1}, x_0)\, q(x_{t-1} \mid x_0)}{q(x_t \mid x_0)}$$

$$= \frac{q(x_t \mid x_{t-1})\, q(x_{t-1} \mid x_0)}{q(x_t \mid x_0)}$$

Diffusion Loss and Noise Parameterization

- **Posterior Conditioned on x_0**



$$q(x_{t-1} \mid x_t, x_0) = \frac{q(x_t, x_{t-1} \mid x_0)}{q(x_t \mid x_0)} = \frac{q(x_t \mid x_{t-1}, x_0)\, q(x_{t-1} \mid x_0)}{q(x_t \mid x_0)}$$

$$= \boxed{\frac{q(x_t \mid x_{t-1})\, q(x_{t-1} \mid x_0)}{q(x_t \mid x_0)}}$$

All Components are Tractable!

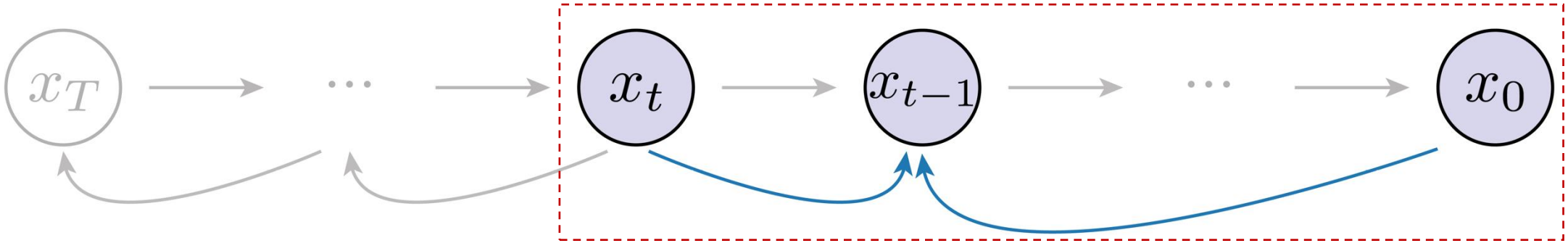
Diffusion Loss and Noise Parameterization

- **Re-visit the Diffusion Loss**

$$L_t = D_{KL}\Big(\big(q(x_{t-1} \mid x_t, x_0) \,\|\, p_\theta(x_{t-1} \mid x_t)\big)\Big)$$







$$\mathbb{E}_q\left[D_{\mathrm{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \,\|\, p(\mathbf{x}_T)) + \sum_{t>1} D_{\mathrm{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \,\|\, p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)) - \log p_\theta(\mathbf{x}_0|\mathbf{x}_1)\right]$$

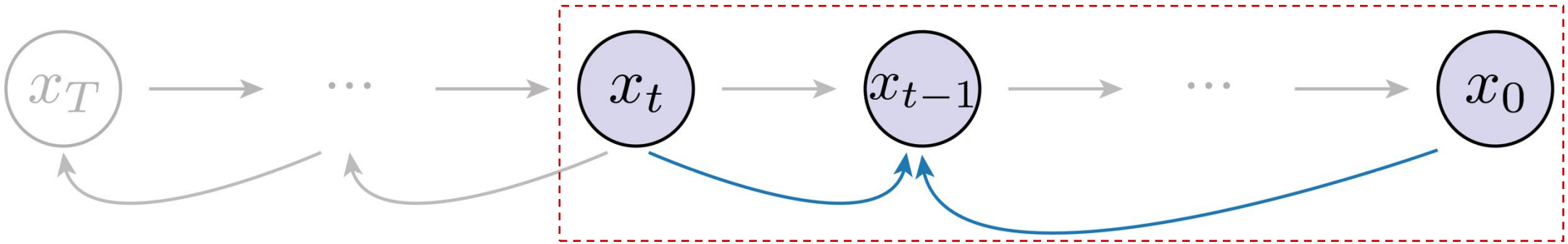
Diffusion Loss and Noise Parameterization

- **Re-visit the Diffusion Loss**



$$L_t = D_{KL}\Big(\big(q(x_{t-1} \mid x_t, x_0) \,\|\, p_\theta(x_{t-1} \mid x_t)\big)\Big)$$





$$\mathbb{E}_q\left[D_{\mathrm{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \parallel p(\mathbf{x}_T)) + \sum_{t>1} D_{\mathrm{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)) - \log p_\theta(\mathbf{x}_0|\mathbf{x}_1)\right]$$

How to implement in a Diffusion Model with Gaussian Noise?

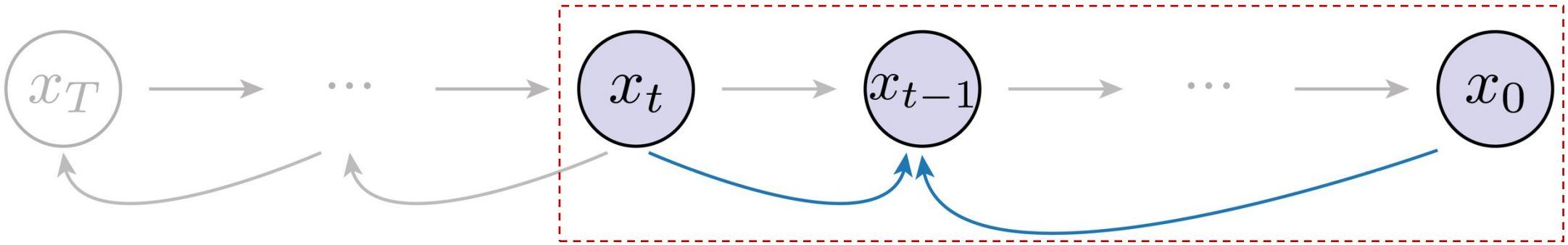
Diffusion Loss and Noise Parameterization

- **Re-visit the Diffusion Loss**



$$L_t = D_{KL}\Big(\big(q(x_{t-1} \mid x_t, x_0) \,\|\, p_\theta(x_{t-1} \mid x_t)\big)\Big)$$





$$q(x_{t-1} \mid x_t, x_0) = \mathcal{N}\big(x_{t-1}; \tilde{\mu}(x_t, x_0), \tilde{\beta}_t I\big)$$

Posterior Induced

by the Forward Process

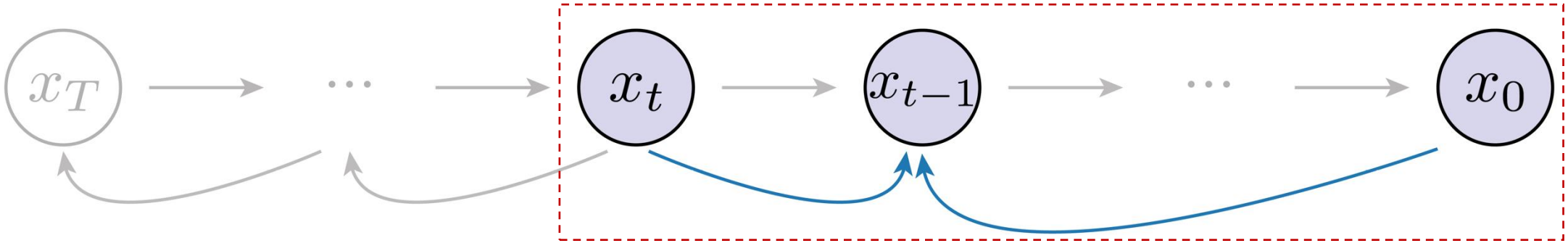
Diffusion Loss and Noise Parameterization

- **Re-visit the Diffusion Loss**



$$L_t = D_{KL}\Big(\big(q(x_{t-1} \mid x_t, x_0) \,\|\, p_\theta(x_{t-1} \mid x_t)\big)\Big)$$





$q(x_{t-1} \mid x_t, x_0) = \mathcal{N}\big(x_{t-1}; \tilde{\mu}(x_t, x_0), \tilde{\beta}_t I\big)$ ⟸ All Components are Gaussian

Posterior Induced by the Forward Process

$$q(x_{t-1} \mid x_t, x_0) = \frac{q(x_t \mid x_{t-1})\, q(x_{t-1} \mid x_0)}{q(x_t \mid x_0)}$$

$q(x_{t-1} \mid x_t, x_0)$ can be expressed **analytically**

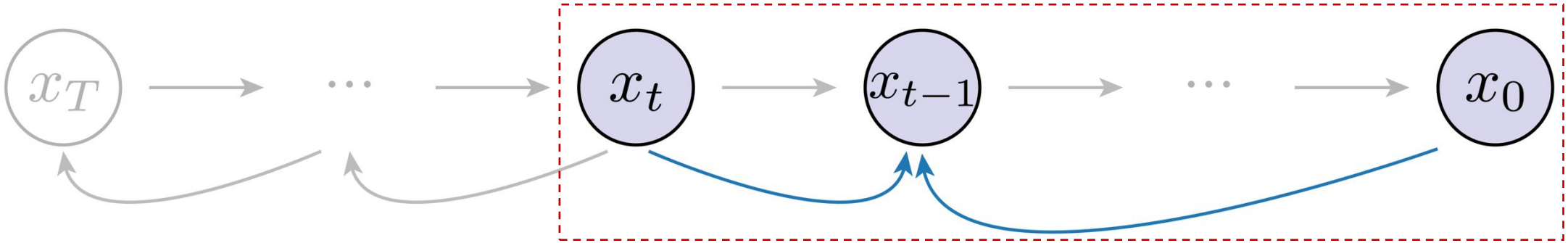
Diffusion Loss and Noise Parameterization

- **Re-visit the Diffusion Loss**



$$L_t = D_{KL}\Big(\big(q(x_{t-1} \mid x_t, x_0) \,\|\, p_\theta(x_{t-1} \mid x_t)\big)\Big)$$





$$q(x_{t-1} \mid x_t, x_0) = \mathcal{N}\big(x_{t-1}; \tilde{\mu}(x_t, x_0), \tilde{\beta}_t I\big)$$

Posterior **Induced** by the Forward Process

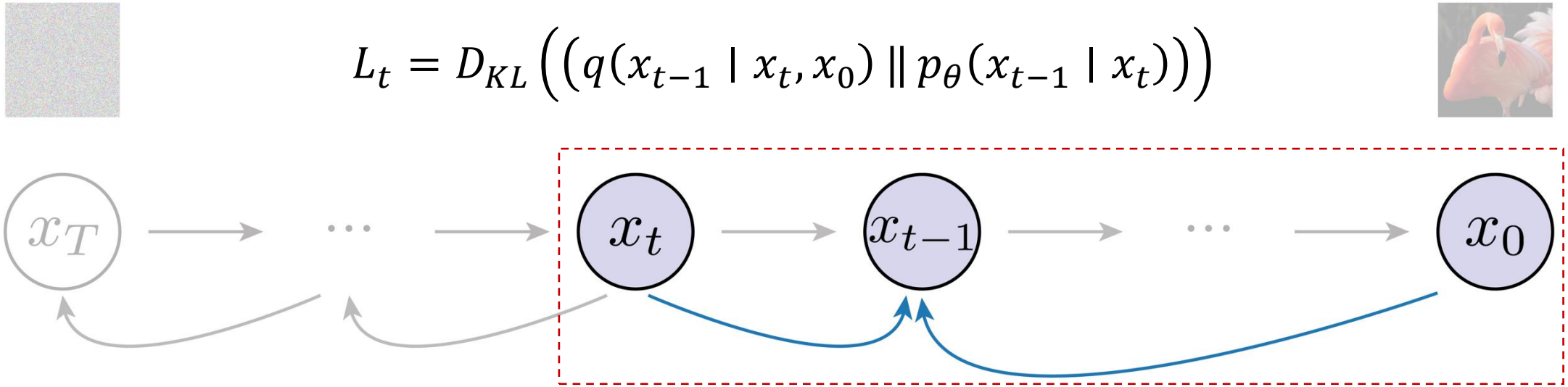
$\Longleftrightarrow$

$$p_\theta(x_{t-1} \mid x_t) = \mathcal{N}\big(x_{t-1}; \mu_\theta(x_t, t), \tilde{\beta}_t I\big)$$

Posterior **Parameterized** by the Neural Network

Diffusion Loss and Noise Parameterization

- **Re-visit the Diffusion Loss**



$$q(x_{t-1} \mid x_t, x_0) = \mathcal{N}\left(x_{t-1}; \tilde{\boldsymbol{\mu}}(\boldsymbol{x_t}, \boldsymbol{x_0}), \tilde{\beta}_t I\right) \iff p_\theta(x_{t-1} \mid x_t) = \mathcal{N}\left(x_{t-1}; \boldsymbol{\mu_\theta}(\boldsymbol{x_t}, \boldsymbol{t}), \tilde{\beta}_t I\right)$$

"Mean-parameterization" with fixed variance

(other distribution modeling methods are also valid)

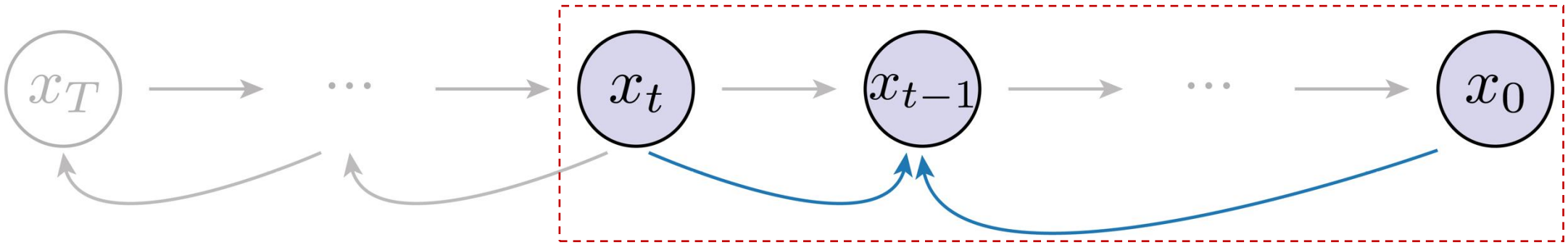
Diffusion Loss and Noise Parameterization

- **Re-visit the Diffusion Loss**



$$L_t = D_{KL}\Big(\big(q(x_{t-1} \mid x_t, x_0) \,\|\, p_\theta(x_{t-1} \mid x_t)\big)\Big)$$





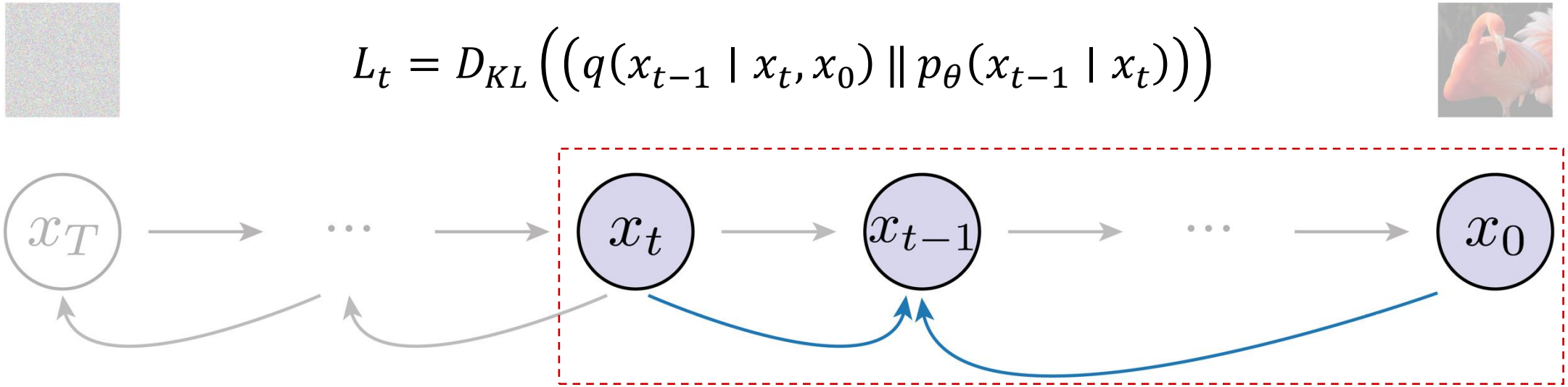
$$q(x_{t-1} \mid x_t, x_0) = \mathcal{N}\big(x_{t-1}; \tilde{\boldsymbol{\mu}}(\boldsymbol{x_t}, \boldsymbol{x_0}), \tilde{\beta}_t I\big) \iff p_\theta(x_{t-1} \mid x_t) = \mathcal{N}\big(x_{t-1}; \boldsymbol{\mu_\theta}(\boldsymbol{x_t}, \boldsymbol{t}), \tilde{\beta}_t I\big)$$

"Mean-parameterization" with fixed variance

Minimizing L_t ➔ Fitting the mean value: $\boldsymbol{\mu_\theta}(\boldsymbol{x_t}, \boldsymbol{t}) \to \tilde{\boldsymbol{\mu}}(\boldsymbol{x_t}, \boldsymbol{x_0})$

Diffusion Loss and Noise Parameterization

- **Re-visit the Diffusion Loss**



$$q(x_{t-1} \mid x_t, x_0) = \mathcal{N}\left(x_{t-1}; \tilde{\mu}(x_t, x_0), \tilde{\boldsymbol{\beta}}_t \boldsymbol{I}\right) \quad \Longleftrightarrow \quad p_\theta(x_{t-1} \mid x_t) = \mathcal{N}\left(x_{t-1}; \mu_\theta(x_t, t), \tilde{\boldsymbol{\beta}}_t \boldsymbol{I}\right)$$

- **Variance of Posterior**

  Only relies on the noise schedule β_t and time step t $\quad \tilde{\beta}_t = \dfrac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t}(1 - \alpha_t)$

Diffusion Loss and Noise Parameterization

- **Noise-Parameterization**

$$q(x_{t-1} \mid x_t, x_0) = \mathcal{N}\left(x_{t-1}; \tilde{\mu}(x_t, x_0), \tilde{\beta}_t I\right)$$

  - One can derive $\tilde{\mu}_t := \tilde{\mu}(x_t, x_0)$ as:

$$\tilde{\mu}_t = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_t\right) \leftarrow \mu_\theta(x_t, t)$$

Ho et al., "Denoising Diffusion Probabilistic Models," NeurIPS 2020.

Diffusion Loss and Noise Parameterization

- **Noise-Parameterization**

$$q(x_{t-1} \mid x_t, x_0) = \mathcal{N}\left(x_{t-1}; \tilde{\mu}(x_t, x_0), \tilde{\beta}_t I\right)$$

  - One can derive $\tilde{\mu}_t := \tilde{\mu}(x_t, x_0)$ as:

$$\tilde{\mu}_t = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_t\right)$$

  - We can instead set our model as

$$\mu_\theta(x_t, t) = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\boldsymbol{\epsilon_\theta(x_t, t)}\right)$$

  - With this "noise-parameterization", minimizing L_t means to **remove the predicted noise ϵ_θ from the corrupted data x_t**

Ho et al., "Denoising Diffusion Probabilistic Models," NeurIPS 2020.

Diffusion Loss and Noise Parameterization

- **Noise Matching**

$$L_t = D_{KL}\Big(\big(q(x_{t-1} \mid x_t, x_0) \,\|\, p_\theta(x_{t-1} \mid x_t)\big)\Big)$$

Derivation

Modeling

$$q(x_{t-1} \mid x_t, x_0) = \mathcal{N}\big(x_{t-1}; \boldsymbol{\tilde{\mu}(x_t, x_0)}, \tilde{\beta}_t I\big)$$

$$p_\theta(x_{t-1} \mid x_t) = \mathcal{N}\big(x_{t-1}; \boldsymbol{\mu_\theta(x_t, t)}, \tilde{\beta}_t I\big)$$

Mean-parameterization

$$\boldsymbol{\tilde{\mu}_t} := \boldsymbol{\tilde{\mu}(x_t, x_0)} = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_t\right) \quad \longleftarrow \quad \boldsymbol{\mu_\theta(x_t, t)}$$

Ho et al., "Denoising Diffusion Probabilistic Models," NeurIPS 2020.

Diffusion Loss and Noise Parameterization

- **Noise Matching**

$$L_t = D_{KL}\Big(\big(q(x_{t-1} \mid x_t, x_0) \,\|\, p_\theta(x_{t-1} \mid x_t)\big)\Big)$$

Derivation

Modeling

$$q(x_{t-1} \mid x_t, x_0) = \mathcal{N}\big(x_{t-1}; \boldsymbol{\tilde{\mu}(x_t, x_0)}, \tilde{\beta}_t I\big)$$

$$p_\theta(x_{t-1} \mid x_t) = \mathcal{N}\big(x_{t-1}; \boldsymbol{\mu_\theta(x_t, t)}, \tilde{\beta}_t I\big)$$

$$\boldsymbol{\tilde{\mu}_t} \coloneqq \boldsymbol{\tilde{\mu}(x_t, x_0)} = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\boldsymbol{\epsilon_t}\right) \Longleftrightarrow \boldsymbol{\mu_\theta(x_t, t)} = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\boldsymbol{\epsilon_\theta(x_t, t)}\right)$$

$$\boldsymbol{\epsilon_t} \Longleftrightarrow \boldsymbol{\epsilon_\theta(x_t, t)}$$

Noise-parameterization

Ho et al., "Denoising Diffusion Probabilistic Models," NeurIPS 2020.

Diffusion Loss and Noise Parameterization

- **Noise Matching**

$$L_t = D_{KL}\Big(\big(q(x_{t-1} \mid x_t, x_0) \,\|\, p_\theta(x_{t-1} \mid x_t)\big)\Big)$$

$$q(x_{t-1} \mid x_t, x_0) = \mathcal{N}\big(x_{t-1}; \tilde{\mu}(x_t, x_0), \tilde{\beta}_t I\big) \qquad p_\theta(x_{t-1} \mid x_t) = \mathcal{N}\big(x_{t-1}; \mu_\theta(x_t, t), \tilde{\beta}_t I\big)$$

➢ For a certain time step t

$$L_t = \mathbb{E}_{\mathbf{x}_0,\epsilon}\left[C_1 \cdot \|\tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0) - \mu_\theta(\mathbf{x}_t, t)\|^2\right]$$

$$= \mathbb{E}_{\mathbf{x}_0,\epsilon}\left[C_2 \cdot \|\epsilon - \epsilon_\theta(\mathbf{x}_t, t)\|^2\right]$$

$$= \mathbb{E}_{\mathbf{x}_0,\epsilon}\left[C_2 \cdot \|\epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, t)\|^2\right]$$

$$C_1 = \frac{1}{2\tilde{\beta}}$$

$$C_2 = \frac{\beta_t^2}{2\tilde{\beta}_t \alpha_t (1-\bar{\alpha}_t)}$$

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon$$

Ho et al., "Denoising Diffusion Probabilistic Models," NeurIPS 2020.

Diffusion Loss and Noise Parameterization

- **Noise Matching**

$$L_t = \mathbb{E}_{\mathbf{x}_0,\epsilon}\left[C_1 \cdot \|\tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0) - \mu_\theta(\mathbf{x}_t, t)\|^2\right]$$

$$= \mathbb{E}_{\mathbf{x}_0,\epsilon}\left[C_2 \cdot \|\epsilon - \epsilon_\theta(\mathbf{x}_t, t)\|^2\right]$$

$$= \mathbb{E}_{\mathbf{x}_0,\epsilon}\left[C_2 \cdot \|\epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, t)\|^2\right]$$

$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon$$

- Optimization for the diffusion loss can be done as follows
  - ① Sample a timestep t uniformly from 1, 2, …, T
  - ② Sample a datapoint x_0
  - ③ Sample noise $\epsilon \sim \mathcal{N}(0, I)$, and generate $x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon$
  - ④ Take a gradient step on $C_2 \cdot \|\epsilon_t - \epsilon_\theta(x_t, t)\|^2$

Diffusion Loss and Noise Parameterization

- **Training and Sampling in DDPM (Ho et al., 2020)**

Algorithm 1 Training

1: **repeat**
2: $\mathbf{x}_0 \sim q(\mathbf{x}_0)$
3: $t \sim \text{Uniform}(\{1, \dots, T\})$
4: $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
5: Take gradient descent step on
$$\nabla_\theta \left\| \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, t) \right\|^2$$
6: **until** converged

➢ Weights of $\|\epsilon_t - \epsilon_\theta(x_t, t)\|^2$ set as 1

Objective	IS	FID
L, learned diagonal $\boldsymbol{\Sigma}$	–	–
L, fixed isotropic $\boldsymbol{\Sigma}$	7.67±0.13	13.51
$\|\tilde{\boldsymbol{\epsilon}} - \boldsymbol{\epsilon}_\theta\|^2$ (L_{simple})	**9.46±0.11**	**3.17**

$$\mathcal{L} = \mathbb{E}_{x_0,t,\epsilon}[C_2 \cdot \|\epsilon - \epsilon_\theta(x_t, t)\|^2]$$

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{x_0,t,\epsilon}[\|\epsilon - \epsilon_\theta(x_t, t)\|^2]$$

Ho et al., "Denoising Diffusion Probabilistic Models," NeurIPS 2020.

Diffusion Loss and Noise Parameterization

- **Training and Sampling in DDPM (Ho et al., 2020)**

Algorithm 1 Training

1: **repeat**
2: $\mathbf{x}_0 \sim q(\mathbf{x}_0)$
3: $t \sim \text{Uniform}(\{1, \dots, T\})$
4: $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
5: Take gradient descent step on
$$\nabla_\theta \left\| \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, t) \right\|^2$$
6: **until** converged

Algorithm 2 Sampling

1: $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
2: **for** $t = T, \dots, 1$ **do**
3: $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ if $t > 1$, else $\mathbf{z} = \mathbf{0}$
4: $\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)\right) + \sigma_t \mathbf{z}$
5: **end for**
6: **return** $\mathbf{x}_0$

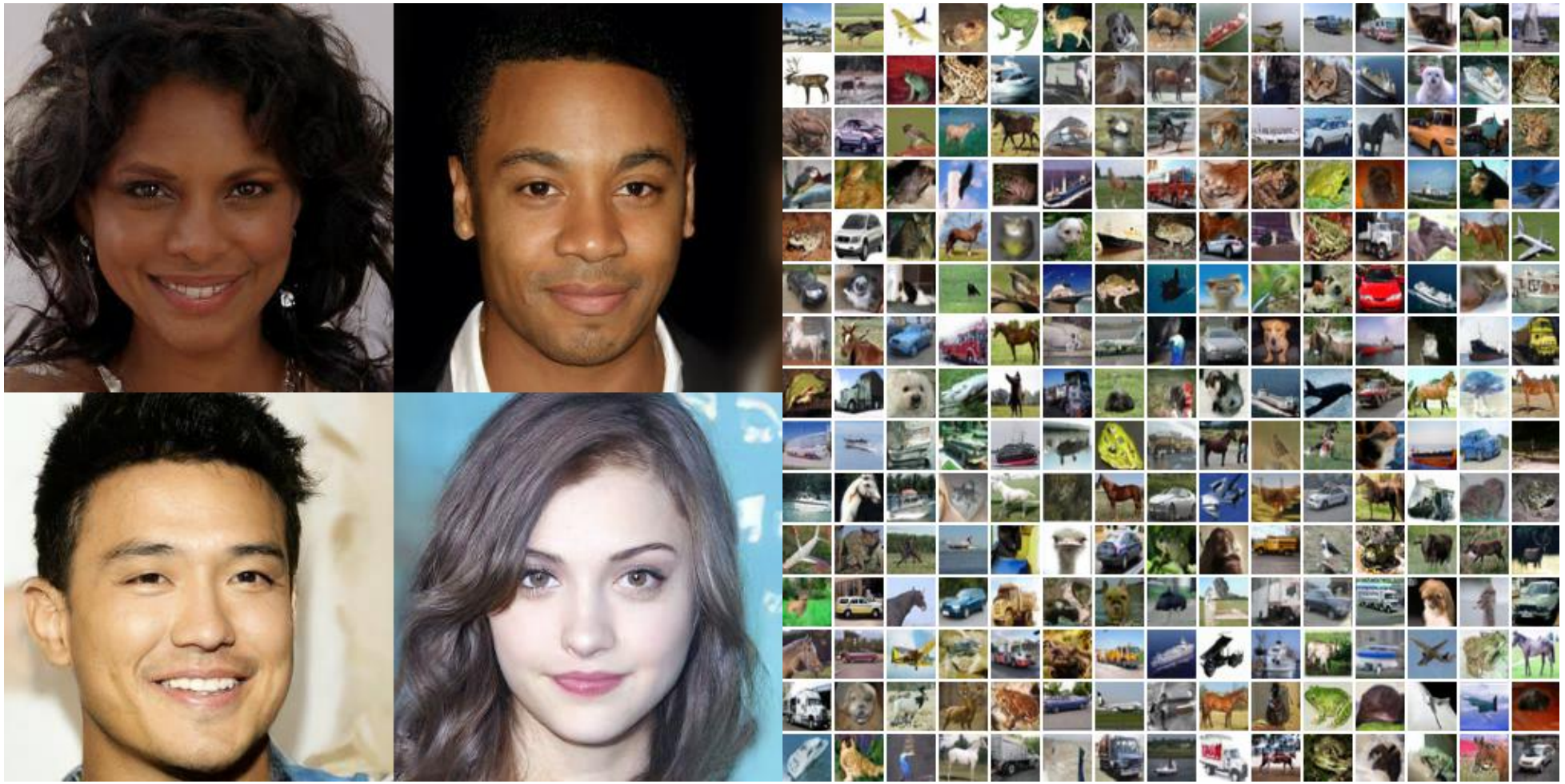
Sampling from estimated Gaussian
$\mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \sigma_t^2 I)$

- "Ancestral Sampling" (祖先采样)
- $\frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_\theta(x_t, t)\right)$ is $\mu_\theta(x_t, t)$, mean of $p_\theta(x_{t-1} \mid x_t)$
- We set $\sigma_t = \sqrt{\tilde{\beta}_t}$, the standard deviation

Ho et al., "Denoising Diffusion Probabilistic Models," NeurIPS 2020.

Diffusion Loss and Noise Parameterization

- **Generation Results of DDPM (Ho et al., 2020)**



Ho et al., "Denoising Diffusion Probabilistic Models," NeurIPS 2020.

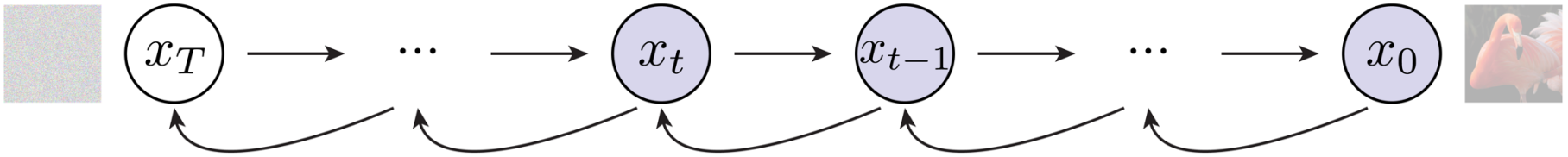


Lecture Outline

- Introduction to Diffusion Models
- Learning Diffusion Models
- Diffusion Loss and Noise Parameterization
- Discussion

Discussion

- **Diffusion Models as Hierarchical VAEs**
  - Diffusion model is a VAE with many noising latents



$$\mathbb{E}_q\left[D_{\mathrm{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \parallel p(\mathbf{x}_T)) + \sum_{t>1} D_{\mathrm{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)) - \log p_\theta(\mathbf{x}_0|\mathbf{x}_1)\right]$$

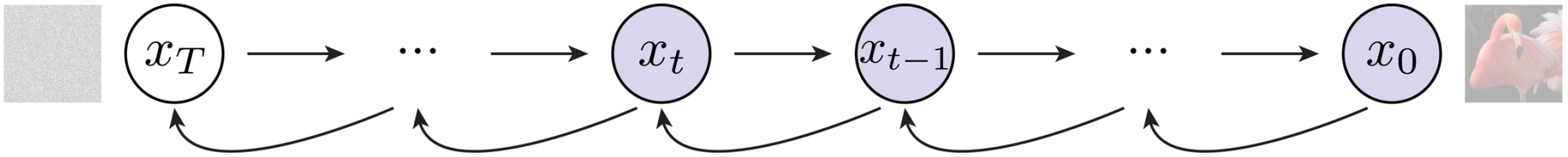
$$L_T = D_{\mathrm{KL}}\big(q(x_T \mid x_0) \parallel p(x_T)\big)$$

$$L_t = D_{KL}\big(q(x_{t-1} \mid x_t, x_0) \parallel p_\theta(x_{t-1} \mid x_t)\big)$$

$$L_0 = -\log p_\theta\,(x_0 \mid x_1)$$

Discussion

- **Diffusion Models as Hierarchical VAEs**
  - ➢ Diffusion model is a VAE with many noising latents



$$\mathbb{E}_q\left[D_{\mathrm{KL}}(q(\mathbf{x}_T|\mathbf{x}_0)\parallel p(\mathbf{x}_T)) + \sum_{t>1} D_{\mathrm{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)\parallel p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)) - \log p_\theta(\mathbf{x}_0|\mathbf{x}_1)\right]$$

- **Latent-variable Model**
- **Similar to ELBO**

$$L_T = D_{\mathrm{KL}}\big(q(x_T \mid x_0) \parallel p(x_T)\big)$$

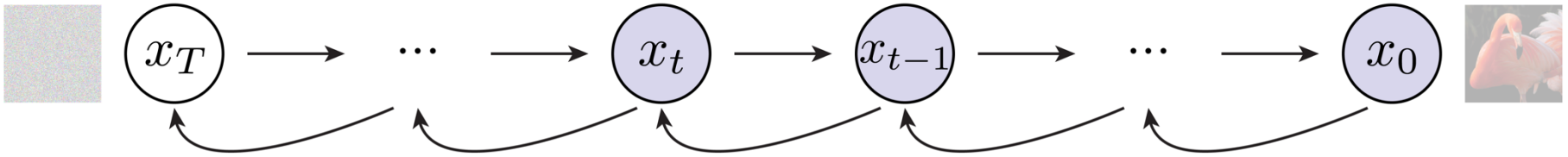
Variational Lower Bound

$$L_t = D_{KL}\big(q(x_{t-1} \mid x_t, x_0) \parallel p_\theta(x_{t-1} \mid x_t)\big)$$

$$L_0 = -\log p_\theta\,(x_0 \mid x_1)$$

Discussion

- **Diffusion Models as Hierarchical VAEs**
  - ➢ Diffusion model is a VAE with many noising latents



$$\mathbb{E}_q\left[D_{\mathrm{KL}}(q(\mathbf{x}_T|\mathbf{x}_0)\parallel p(\mathbf{x}_T)) + \sum_{t>1} D_{\mathrm{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)\parallel p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)) - \log p_\theta(\mathbf{x}_0|\mathbf{x}_1)\right]$$

$$L_T = D_{\mathrm{KL}}\big(q(z \mid x_0) \parallel p(z)\big)$$

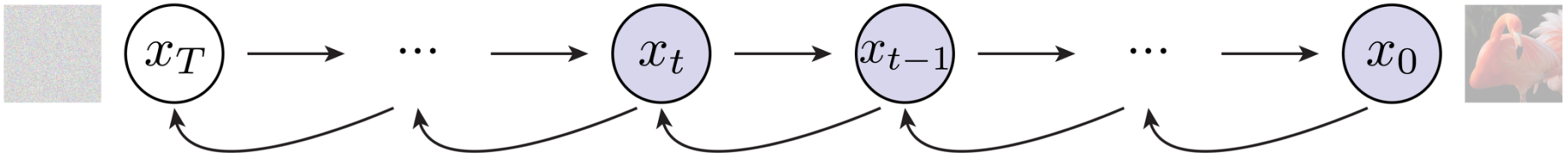
~~$$L_t = D_{KL}\big(q(x_{t-1} \mid x_t, x_0) \parallel p_\theta(x_{t-1} \mid x_t)\big)$$~~

It's ELBO if one step !

$$L_0 = -\log p_\theta\,(x_0 \mid z)$$

Discussion

- **Diffusion Models as Hierarchical VAEs**
  - Diffusion model is a VAE with many noising latents



$$\mathbb{E}_q\left[D_{\mathrm{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \parallel p(\mathbf{x}_T)) + \sum_{t>1} D_{\mathrm{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)) - \log p_\theta(\mathbf{x}_0|\mathbf{x}_1)\right]$$

$$L_T = D_{\mathrm{KL}}\big(q(x_T \mid x_0) \parallel p(x_T)\big)$$

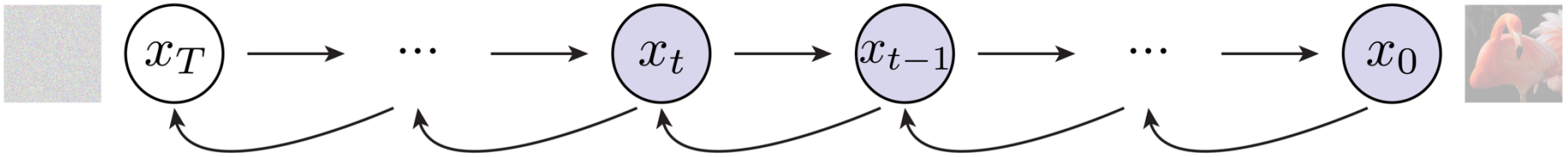
$$L_t = D_{KL}\big(q(x_{t-1} \mid x_t, x_0) \parallel p_\theta(x_{t-1} \mid x_t)\big)$$

$$L_0 = -\log p_\theta\,(x_0 \mid x_1)$$

- **Constant if $p(x_T)$ set to Gaussian**
- **Unlike VAE, q is not parameterized**

Discussion

- **Diffusion Models as Hierarchical VAEs**
  - Diffusion model is a VAE with many noising latents



$$\mathbb{E}_q\left[D_{\mathrm{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \parallel p(\mathbf{x}_T)) + \sum_{t>1} D_{\mathrm{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)) - \log p_\theta(\mathbf{x}_0|\mathbf{x}_1)\right]$$

$$L_T = D_{\mathrm{KL}}\big(q(x_T \mid x_0) \parallel p(x_T)\big)$$

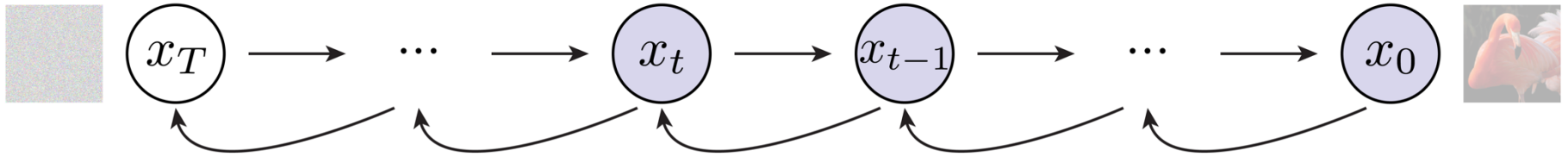
$$L_t = D_{KL}\big(q(x_{t-1} \mid x_t, x_0) \parallel p_\theta(x_{t-1} \mid x_t)\big)$$

$$L_0 = -\log p_\theta\,(x_0 \mid x_1)$$

Reconstruction loss, similar to VAE

Discussion

- **Diffusion Models as Hierarchical VAEs**
  - Diffusion model is a VAE with many noising latents



$$\mathbb{E}_q\left[D_{\mathrm{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \parallel p(\mathbf{x}_T)) + \sum_{t>1} D_{\mathrm{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)) - \log p_\theta(\mathbf{x}_0|\mathbf{x}_1)\right]$$

$$L_T = D_{\mathrm{KL}}\big(q(x_T \mid x_0) \parallel p(x_T)\big)$$

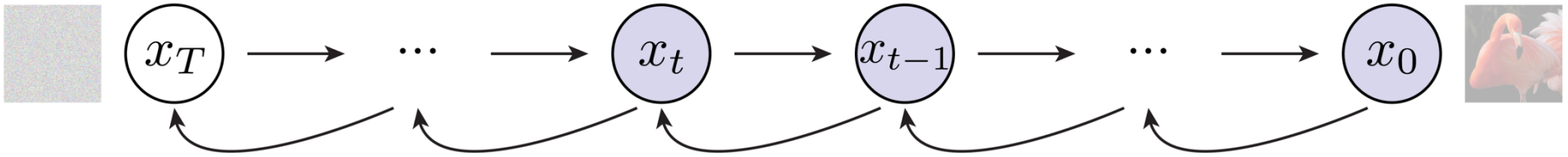
$$L_t = D_{KL}\big(q(x_{t-1} \mid x_t, x_0) \parallel p_\theta(x_{t-1} \mid x_t)\big)$$

$$L_0 = -\log p_\theta\,(x_0 \mid x_1)$$

- **KL Divergence for intermediate layers**
- **L2 loss on noise**

Discussion

- **Diffusion Models as Hierarchical VAEs**
  - ➢ Diffusion model is a VAE with many noising latents

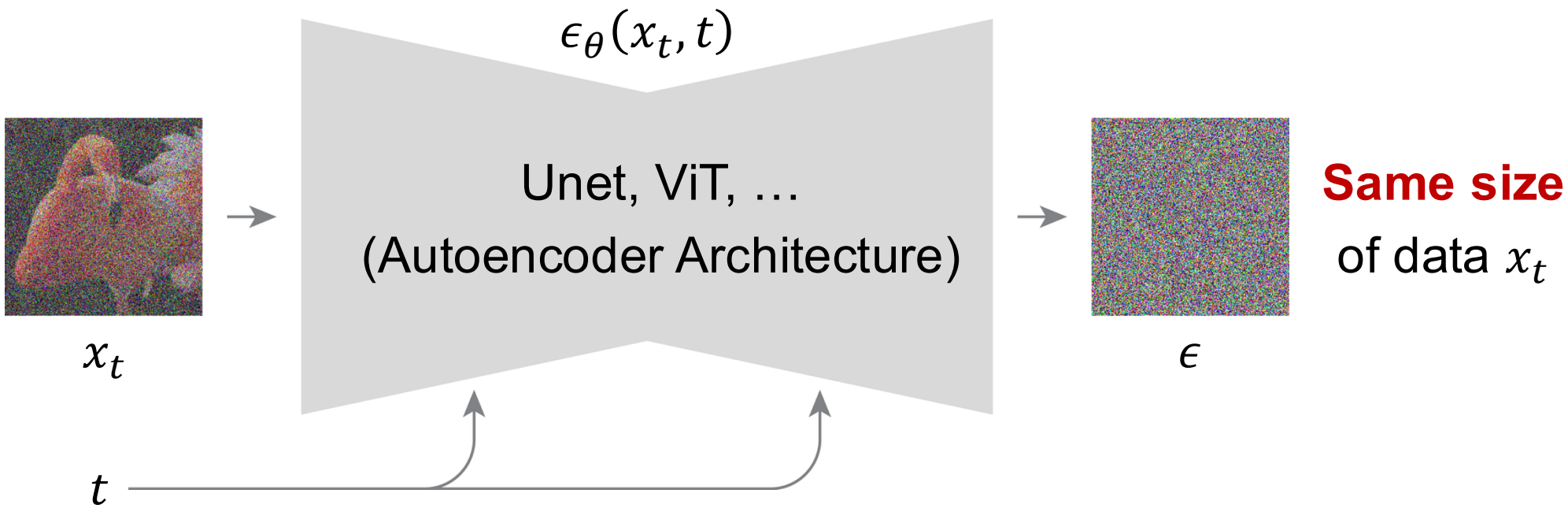


$$\mathbb{E}_q \left[D_{\mathrm{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \parallel p(\mathbf{x}_T)) + \sum_{t>1} D_{\mathrm{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)) - \log p_\theta(\mathbf{x}_0|\mathbf{x}_1) \right]$$

  - ➢ Diffusion models decompose a distribution into **many** simpler ones
    - ✓ We need T models to fit $p_\theta(x_{t-1} \mid x_t)$ for all $t = 1, \ldots, T$
    - ✓ Or we can use a **single network conditioned on the noise level t**

Discussion

- **Noise Conditional Network**
  - ➢ Using noise-parameterization, the model $\epsilon_\theta(x_t, t)$ is conditioned on:
    - ① **noisy image x_t**
    - ② **time step t**



Discussion

- **Clean Data Estimation**
  - **Note:** Given the following facts

    ① **Forward process** $x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon$

    ② **Noise Prediction Model** $\epsilon_\theta(x_t, t)$

    one can **estimate the clean data from any noisy state x_t**

$$\hat{x}_0 = \frac{x_t - \sqrt{1 - \bar{\alpha}_t}\,\epsilon_\theta(x_t, t)}{\sqrt{\bar{\alpha}_t}}$$

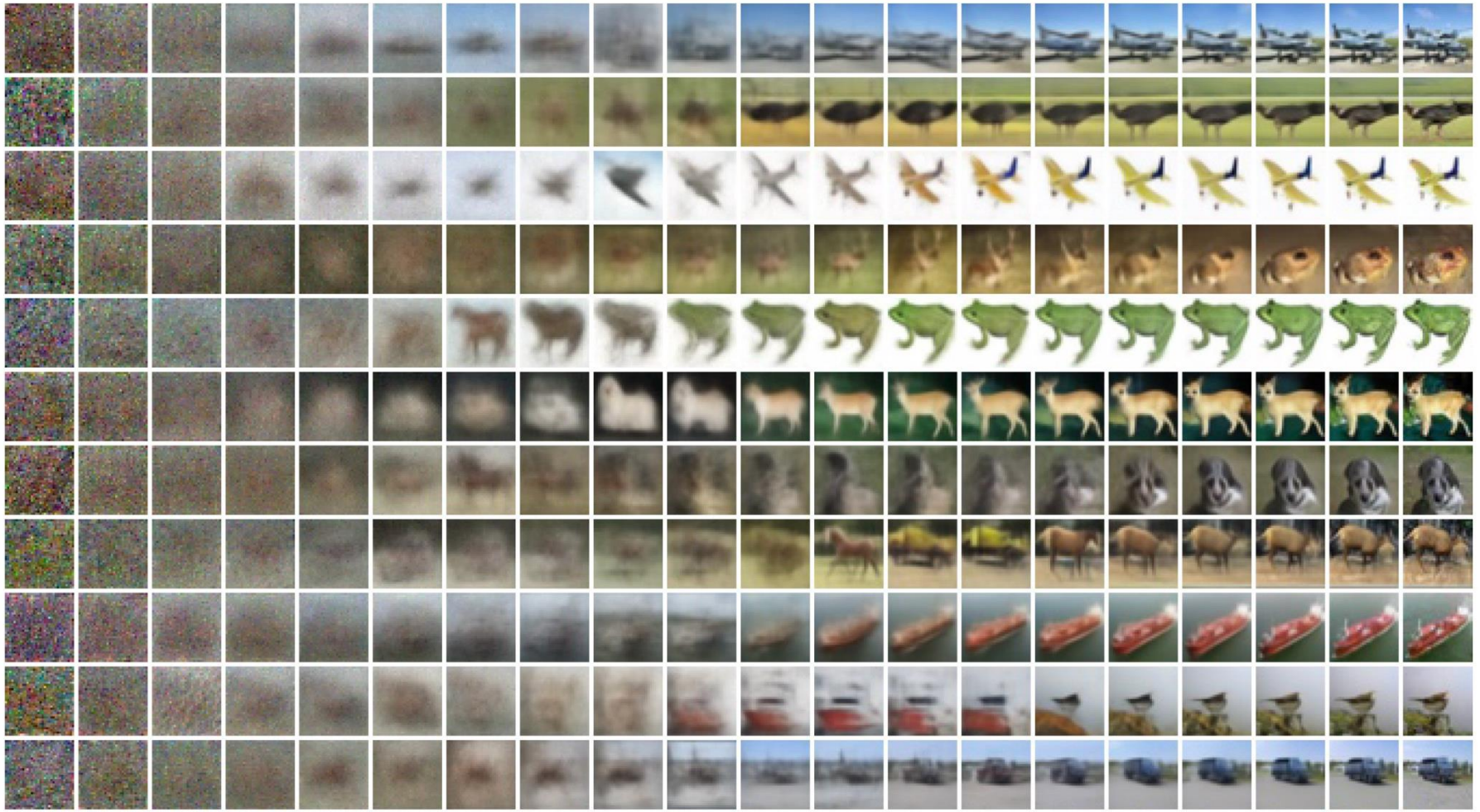


Current guess of clean x_0 at time step t

Discussion

- **Clean Data Estimation in Progression**

$t = T$



$t = 0$

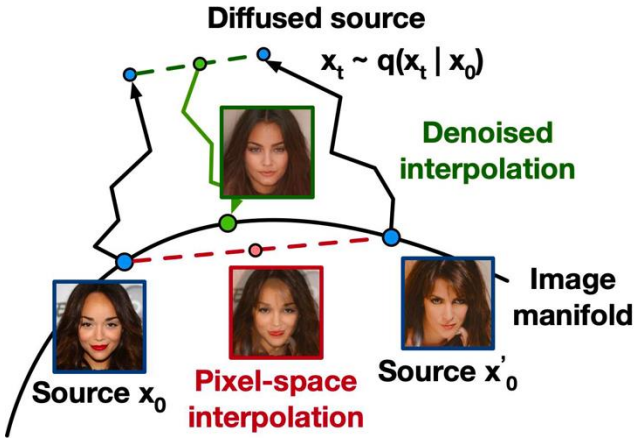
$\hat{x}_0(t)$

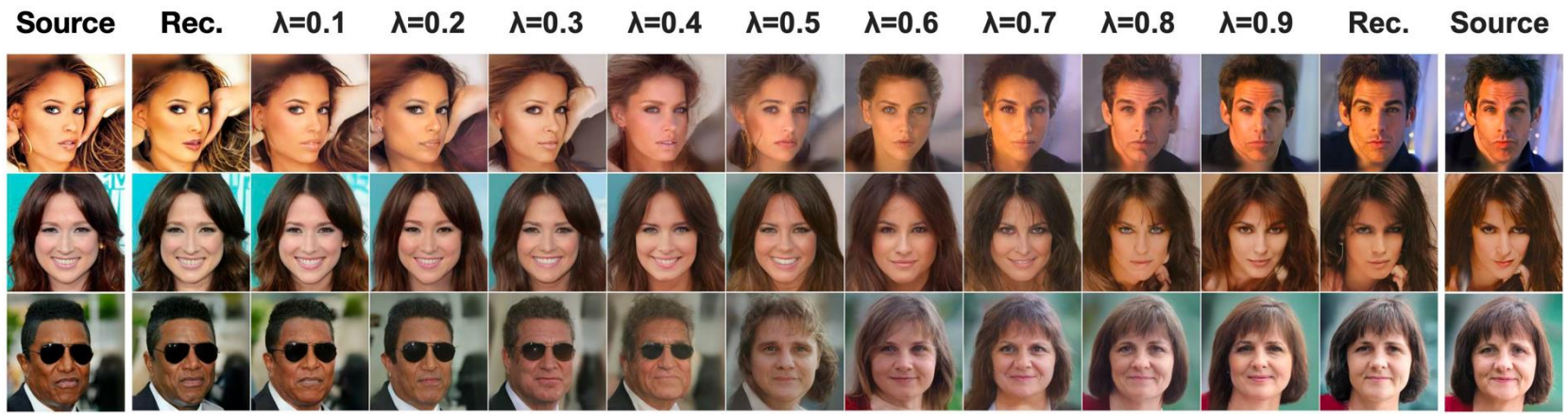
Ho et al., “Denoising Diffusion Probabilistic Models,” NeurIPS 2020.

Discussion

- **Latent States of Diffusion Models**



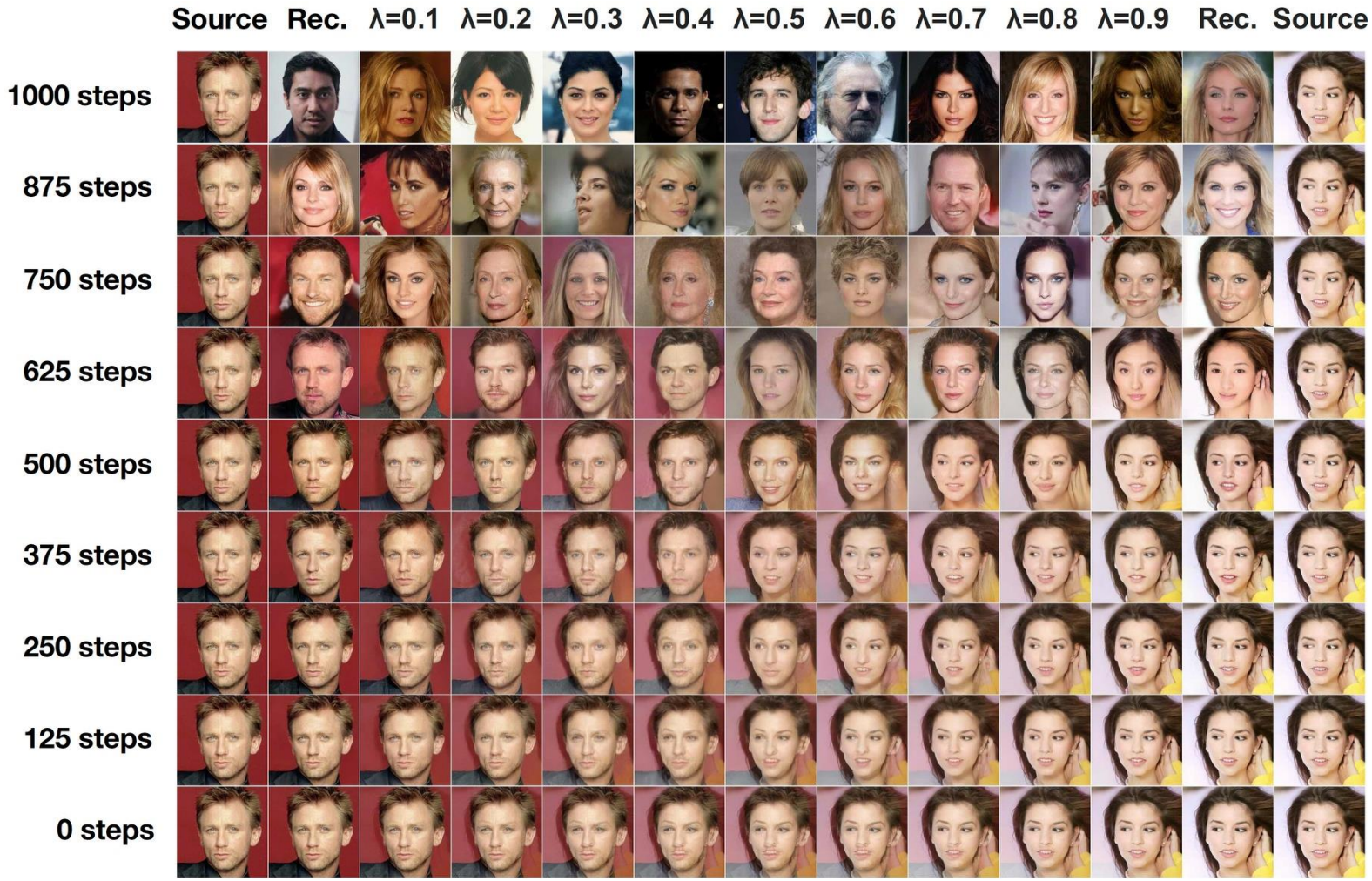




Ho et al., "Denoising Diffusion Probabilistic Models," NeurIPS 2020.

Discussion

- **Latent States of Diffusion Models**



Interpolation with varying t

Ho et al., "Denoising Diffusion Probabilistic Models," NeurIPS 2020.



Content

- Introduction to Diffusion Models
- Learning Diffusion Models
- Diffusion Loss and Noise Parameterization
- Discussion

Summary

- **Pros**
  - ➢ High-quality samples with stable training objective
  - ➢ Less prone to mode collapse than GANs
  - ➢ Effective conditional and controllable generation (following lectures)
  - ➢ Natural connection to score matching and flow models (following lectures)
- **Cons**
  - ➢ Slow sampling due to iterative denoising
  - ➢ High computational cost
  - ➢ Less straightforward to discrete data

    ➔ Active research directions!

Supplementary Material

- **Cornell CS 6785 Lecture 13**
  - https://www.youtube.com/watch?v=KQOQyrkU8rk&list=PL2UML_KCiC0UPzjW9BjO-IW6dqliu9O4B&index=13
- **Deep Generative Models**
  - Chapter 5.5 Hierarchical Latent Variable Models
- **The Principles of Diffusion Models**
  - Chapter B.2 Variational Perspective: From VAEs to DDPMs
- **Lil's Blog**
  - What are Diffusion Models?
  - https://lilianweng.github.io/posts/2021-07-11-diffusion-models/

Readling List

- **Must-read Material**

- Sohl-Dickstein et al., “Deep Unsupervised Learning using Nonequilibrium Thermodynamics,” **ICML 2015**
- Song and Ermon, “Generative Modeling by Estimating Gradients of the Data Distribution,” **NeurIPS 2019**
- Ho et al., “Denoising Diffusion Probabilistic Models,” **NeurIPS 2020**
- Song et al., “Denoising Diffusion Implicit Models,” **ICLR 2021**
- Song et al., “Score-Based Generative Modeling through Stochastic Differential Equations,” **ICLR 2021**
- Nichol and Dhariwal, “Improved Denoising Diffusion Probabilistic Models,” **ICML 2021**
- Dhariwal and Nichol, “Diffusion Models Beat GANs on Image Synthesis,” **NeurIPS 2021**
- Ho and Salimans, “Classifier-Free Diffusion Guidance,” **NeurIPS Workshop 2021**
- Rombach et al., “High-Resolution Image Synthesis with Latent Diffusion Models,” **CVPR 2022**
- Karras et al., “Elucidating the Design Space of Diffusion-Based Generative Models,” **NeurIPS 2022**
- **Dieleman, “Perspectives on diffusion”. https://sander.ai/2023/07/20/perspectives.html**